第四章　積分

4.1 不定積分

定義 (反導函數)**.** 給定 $F(x)$, 若 $\dfrac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x)$, 則稱 $F(x)$ 為 $f(x)$ 的反導函數 (antiderivative) .

性質. 若 $F(x)$, $G(x)$ 分別為 $f(x)$, $g(x)$ 的反導函數, $c \in \mathbb{R}$. 則

- $F(x) + c$ 為 $f(x)$ 的反導函數.
- $c\,F(x)$ 為 $c\,f(x)$ 的反導函數.
- $F(x) + G(x)$ 為 $f(x) + g(x)$ 的反導函數.

結論.

- $\dfrac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x) \implies \mathrm{d}F(x) = f(x)\cdot\mathrm{d}x \implies F(x) = \displaystyle\int f(x)\cdot\mathrm{d}x = \int f(x)\,\mathrm{d}x$
- $F(x)$ 為 $f(x)$ 的反導函數 $\iff$ $f(x)$ 的反導函數為 $F(x)$ $\iff$ $F(x)$ 的導函數為 $f(x)$ $\iff$ $F(x)$ (對 x) 的微分為 $f(x)$ $\iff$ $f(x)$ (對 x) 的 (不定) 積分為 $F(x)$
- $f(x)$ 的反導函數 $\equiv$ $f(x)$ (對 x) 的 (不定) 積分
- 基礎積分集: 以下 $\alpha \neq -1$, $a \neq 0$.

$f(x)$	x^α	$\dfrac{1}{x}$	e^{ax}	$\sin ax$	$\cos ax$	$\dfrac{1}{\sqrt{a^2-x^2}}$	$\dfrac{1}{a^2+x^2}$
$\displaystyle\int f(x)\,\mathrm{d}x$	$\dfrac{1}{\alpha+1}x^{\alpha+1}$	$\ln\lvert x\rvert$	$\dfrac{1}{a}e^{ax}$	$-\dfrac{1}{a}\cos ax$	$\dfrac{1}{a}\sin ax$	$\sin^{-1}\dfrac{x}{\lvert a\rvert}$	$\dfrac{1}{a}\tan^{-1}\dfrac{x}{a}$

- (Liouville) e^{-x^2}, $\dfrac{e^x}{x}$, $\dfrac{1}{\ln x}$, $\sin(x^2)$, $\cos(x^2)$, $\dfrac{\sin x}{x}$, $\dfrac{\cos x}{x}$, x^x 無 (初等函數形式之) 反導函數!

例.

1. $\displaystyle\int \sqrt{x}\,\mathrm{d}x = \frac{2}{3}x^{\frac{3}{2}} + c$
2. $\displaystyle\int x^\pi\,\mathrm{d}x = \frac{1}{\pi+1}x^{\pi+1} + c$
3. $\displaystyle\int \sin\pi x\,\mathrm{d}x = -\frac{1}{\pi}\cos\pi x + c$
4. $\displaystyle\int \cos xy\,\mathrm{d}x = \frac{1}{y}\sin xy + c$
5. $\displaystyle\int e^{-2x}\,\mathrm{d}x = -\frac{1}{2}e^{-2x} + c$
6. $\displaystyle\int \frac{1}{\sqrt{\pi-x^2}}\,\mathrm{d}x = \sin^{-1}\frac{x}{\sqrt{\pi}} + c$
7. $\displaystyle\int \frac{1}{e+u^2}\,\mathrm{d}u = \frac{1}{\sqrt{e}}\tan^{-1}\frac{u}{\sqrt{e}} + c$
8. $\displaystyle\int \left(\frac{\pi}{x} - e^{\pi x}\right)\mathrm{d}u = \pi\ln x - \frac{e^{\pi x}}{\pi} + c$
9. $\displaystyle\int \frac{3+x^2}{1+x^2}\,\mathrm{d}u = x + 2\tan^{-1}x + c$

習題. 求下列不定積分.

1. $\displaystyle\int \frac{x^3-1}{x^3}\,\mathrm{d}x = x + \frac{1}{2x^3} + c$
2. $\displaystyle\int 5 - \frac{1}{\sqrt{x}}\,\mathrm{d}x = 5x - 2\sqrt{x} + c$
3. $\displaystyle\int (t-1)(t+1)\,\mathrm{d}t = \frac{t^3}{3} - t + c$
4. $\displaystyle\int (\sqrt{x}+1)^2\,\mathrm{d}x = \frac{x^2}{2} + x + \frac{4x^{\frac{3}{2}}}{3} + c$
5. $\displaystyle\int x\sqrt{3x}\,\mathrm{d}x = \frac{2\sqrt{3}}{5}x^{\frac{5}{2}} + c$
6. $\displaystyle\int \frac{1}{x^3} - \frac{1}{x^5}\,\mathrm{d}x = \frac{-1}{2x^2} + \frac{1}{4x^4} + c$
7. $\displaystyle\int \sec^2 x - \sec x\tan x\,\mathrm{d}x = \tan x - \sec x + c$

變數變換法

結論. $\dfrac{\mathrm{d}}{\mathrm{d}x}f(g(x)) = f'(g(x))\cdot g'(x) \implies \mathrm{d}f(g(x)) = f'(g(x))\cdot g'(x)\,\mathrm{d}x \implies f(g(x)) = \displaystyle\int f'(g(x))\cdot g'(x)\,\mathrm{d}x.$
令 $u = g(x)$, 則 $\dfrac{\mathrm{d}u}{\mathrm{d}x} = g'(x) \implies \mathrm{d}u = g'(x)\,\mathrm{d}x$; 故 $\displaystyle\int f'(g(x))\cdot g'(x)\,\mathrm{d}x = \int f'(u)\,\mathrm{d}u = f(u)+c = f(g(x))+c.$

例. 求 $\displaystyle\int \frac{x}{\sqrt{x+1}}\,\mathrm{d}x.$

解.

- (解一) 令 $u = x+1$, 則 $x = u-1$, $\mathrm{d}u = \mathrm{d}x$. 故 $\displaystyle\int \frac{x}{\sqrt{x+1}}\,\mathrm{d}x = \int \frac{u-1}{\sqrt{u}}\,\mathrm{d}u = \int u^{\frac{1}{2}} - u^{-\frac{1}{2}}\,\mathrm{d}u = \frac{2}{3}\,u^{\frac{3}{2}} - 2\sqrt{u} + c = \frac{2}{3}\,(x+1)^{\frac{3}{2}} - 2\sqrt{x+1} + c.$
- (解二) $\displaystyle\int \frac{x}{\sqrt{x+1}}\,\mathrm{d}x = \int \frac{x+1-1}{\sqrt{x+1}}\,\mathrm{d}x = \int \sqrt{x+1} - \frac{1}{\sqrt{x+1}}\,\mathrm{d}x.$ 令 $u = x+1$, 則 $\mathrm{d}u = \mathrm{d}x$. 故 $\displaystyle\int \sqrt{x+1} - \frac{1}{\sqrt{x+1}}\,\mathrm{d}x = \int u^{\frac{1}{2}} - u^{-\frac{1}{2}}\,\mathrm{d}u = \frac{2}{3}\,u^{\frac{3}{2}} - 2\sqrt{u} + c = \frac{2}{3}\,(x+1)^{\frac{3}{2}} - 2\sqrt{x+1} + c.$
- (解三) 令 $u = \sqrt{x+1}$, 則 $x = u^2-1$, $\mathrm{d}u = \dfrac{1}{2\sqrt{x+1}}\,\mathrm{d}x \implies \dfrac{1}{\sqrt{x+1}}\,\mathrm{d}x = 2\,\mathrm{d}u$. 故 $\displaystyle\int \frac{x}{\sqrt{x+1}}\,\mathrm{d}x = \int x\cdot\frac{1}{\sqrt{x+1}}\,\mathrm{d}x = \int (u^2-1)\cdot 2\,\mathrm{d}u = \frac{2}{3}\,u^3 - 2u + c = \frac{2}{3}\,(x+1)^{\frac{3}{2}} - 2\sqrt{x+1} + c.$

例. 求 $\displaystyle\int \frac{x}{x^2+1}\,\mathrm{d}x.$

解. 令 $u = x^2+1$, 則 $\mathrm{d}u = 2\,x\,\mathrm{d}x \implies x\,\mathrm{d}x = \dfrac{1}{2}\,\mathrm{d}u$. 故 $\displaystyle\int \frac{x}{x^2+1}\,\mathrm{d}x = \int \frac{1}{x^2+1}\cdot x\,\mathrm{d}x = \int \frac{1}{u}\cdot\frac{1}{2}\,\mathrm{d}u = \frac{1}{2}\ln|u| + c = \frac{1}{2}\ln(x^2+1) + c.$

例. 求 $\displaystyle\int \frac{\sin(3\ln x)}{x}\,\mathrm{d}x.$

解. 令 $u = \ln x$, 則 $\mathrm{d}u = \dfrac{1}{x}\,\mathrm{d}x$. 故 $\displaystyle\int \frac{\sin(3\ln x)}{x}\,\mathrm{d}x = \int \sin(3\ln x)\cdot\frac{1}{x}\,\mathrm{d}x = \int \sin 3u\cdot\mathrm{d}u = -\frac{1}{3}\cos 3u + c = -\frac{1}{3}\cos(3\ln x) + c.$

例. 求 $\displaystyle\int e^x\sqrt{1+e^x}\,\mathrm{d}x.$

解. 令 $u = 1+e^x$, 則 $\mathrm{d}u = e^x\,\mathrm{d}x$. 故 $\displaystyle\int e^x\sqrt{1+e^x}\,\mathrm{d}x = \int \sqrt{1+e^x}\cdot e^x\,\mathrm{d}x = \int \sqrt{u}\cdot\mathrm{d}u = \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + c = \frac{2}{3}(1+e^x)^{\frac{3}{2}} + c$

例. 求 $\displaystyle\int \frac{e^x}{\sqrt{2-e^{2x}}}\,\mathrm{d}x.$

解. 令 $u = e^x$, 則 $\mathrm{d}u = e^x\,\mathrm{d}x$. 故 $\displaystyle\int \frac{e^x}{\sqrt{2-e^{2x}}}\,\mathrm{d}x = \int \frac{1}{\sqrt{2-u^2}}\,\mathrm{d}u = \sin^{-1}\frac{u}{\sqrt{2}} + c = \sin^{-1}\frac{e^x}{\sqrt{2}} + c.$

例. 求 $\displaystyle\int \frac{1}{\sqrt{e^{2x}-1}}\,\mathrm{d}x.$

解. $\displaystyle\int\frac{1}{\sqrt{e^{2x}-1}}\,\mathrm{d}x=\int\frac{1}{e^x\sqrt{1-e^{-2x}}}\,\mathrm{d}x=\int\frac{e^{-x}}{\sqrt{1-e^{-2x}}}\,\mathrm{d}x$. 令 $u=e^{-x}$, 則 $\mathrm{d}u=-e^{-x}\,\mathrm{d}x\implies e^{-x}\,\mathrm{d}x=-\mathrm{d}u$; 故 $\displaystyle\int\frac{e^{-x}}{\sqrt{1-e^{-2x}}}\,\mathrm{d}x=\int\frac{1}{\sqrt{1-e^{-2x}}}\cdot e^{-x}\,\mathrm{d}x=-\int\frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u=-\sin^{-1}u+c=-\sin^{-1}e^{-x}+c$.

例. 求 $\displaystyle\int\frac{1}{x^2+4x+5}\,\mathrm{d}x$.

解. $\displaystyle\int\frac{1}{x^2+4x+5}\,\mathrm{d}x=\int\frac{1}{(x+2)^2+1}\,\mathrm{d}x$. 令 $u=x+2$, 則 $\mathrm{d}u=\mathrm{d}x$; 故 $\displaystyle\int\frac{1}{(x+2)^2+1}\,\mathrm{d}x=\int\frac{1}{u^2+1}\,\mathrm{d}u=\tan^{-1}u+c=\tan^{-1}(x+2)+c$.

例. 求 $\displaystyle\int\frac{1}{\sqrt{4+2x-x^2}}\,\mathrm{d}x$

解. $\displaystyle\int\frac{1}{\sqrt{4+2x-x^2}}\,\mathrm{d}x=\int\frac{1}{\sqrt{5-(x-1)^2}}\,\mathrm{d}x$. 令 $u=x-1$, 則 $\mathrm{d}u=\mathrm{d}x$; 故 $\displaystyle\int\frac{1}{\sqrt{5-(x-1)^2}}\,\mathrm{d}x=\int\frac{1}{\sqrt{5-u^2}}\,\mathrm{d}u=\sin^{-1}\frac{u}{\sqrt{5}}+c=\sin^{-1}\frac{x-1}{\sqrt{5}}+c$.

例. 求 $\displaystyle\int\tan x\,\mathrm{d}x$.

解. $\displaystyle\int\tan x\,\mathrm{d}x=\int\frac{\sin x}{\cos x}\,\mathrm{d}x$. 令 $u=\cos x$, 則 $\mathrm{d}u=-\sin x\,\mathrm{d}x$; 故 $\displaystyle\int\frac{\sin x}{\cos x}\,\mathrm{d}x=\int\frac{-1}{u}\,\mathrm{d}u=-\ln|u|+c=-\ln|\cos x|+c=\ln|\sec x|+c$

例. 求 $\displaystyle\int\sec x\,\mathrm{d}x$.

解. $\displaystyle\int\sec x\,\mathrm{d}x=\int\frac{\sec x\cdot(\sec x+\tan x)}{\sec x+\tan x}\,\mathrm{d}x$. 令 $u=\sec x+\tan x$, 則 $\mathrm{d}u=(\sec^2x+\sec x\tan x)\,\mathrm{d}x$; 故 $\displaystyle\int\frac{\sec x\cdot(\sec x+\tan x)}{\sec x+\tan x}\,\mathrm{d}x=\int\frac{1}{u}\,\mathrm{d}u=\ln|u|+c=\ln|\sec x+\tan x|+c$

例. 令 $\displaystyle T_1=\int\frac{\sin x}{a\cos x+b\sin x}\,\mathrm{d}x$, $\displaystyle T_2=\int\frac{\cos x}{a\cos x+b\sin x}\,\mathrm{d}x$, a, $b\neq0$, 求 T_1, T_2.

解.

(a) $\displaystyle bT_1+aT_2=\int\frac{b\sin x}{a\cos x+b\sin x}\,\mathrm{d}x+\int\frac{a\cos x}{a\cos x+b\sin x}\,\mathrm{d}x=\int\frac{b\sin x+a\cos x}{a\cos x+b\sin x}\,\mathrm{d}x=\int1\,\mathrm{d}x=x$.

(b) $\displaystyle -aT_1+bT_2=\int\frac{-a\sin x}{a\cos x+b\sin x}\,\mathrm{d}x+\int\frac{b\cos x}{a\cos x+b\sin x}\,\mathrm{d}x=\int\frac{-a\sin x+b\cos x}{a\cos x+b\sin x}\,\mathrm{d}x=\int\frac{\mathrm{d}u}{u}=\ln u=\ln|a\cos x+b\sin x|$ (令 $u=a\cos x+b\sin x$, 則 $\mathrm{d}u=(-a\sin x+b\cos x)\,\mathrm{d}x$) .

解 T_1, T_2 方程式 (a), (b) 得 $\displaystyle T_1=\frac{bx-a\ln|a\cos x+b\sin x|}{a^2+b^2}$, $\displaystyle T_2=\frac{ax+b\ln|a\cos x+b\sin x|}{a^2+b^2}$.

例. 求 $\displaystyle\int\cos^5ax\,\mathrm{d}x$, $a\neq0$.

解. $\displaystyle\int\cos^5ax\,\mathrm{d}x=\int(1-\sin^2ax)^2\cos ax\,\mathrm{d}x$. 令 $u=\sin ax$, 則 $\mathrm{d}u=a\cos ax\,\mathrm{d}x$; 故 $\displaystyle\int(1-\sin^2ax)^2\cos ax\,\mathrm{d}x=\frac{1}{a}\int(1-u^2)^2\,\mathrm{d}u=\frac{1}{a}\int(1-2u^2+u^4)\,\mathrm{d}u=\frac{1}{a}\left(u-\frac{2}{3}u^3+\frac{1}{5}u^5\right)+c=\frac{1}{a}\left(\sin ax-\frac{2}{3}\sin^3ax+\frac{1}{5}\sin^5ax\right)+c$.

例. 求 $\displaystyle\int\cos^4ax\,\mathrm{d}x$, $a\neq0$.

解. $\displaystyle\int \cos^4 ax\,\mathrm{d}x = \int \Big(\frac{1+\cos 2ax}{2}\Big)^2 \mathrm{d}x = \frac{1}{4}\int \big(1+2\cos 2ax+\cos^2 2ax\big)\,\mathrm{d}x = \frac{1}{4}\int \big(1+2\cos 2ax+\frac{1+\cos 4ax}{2}\big)\,\mathrm{d}x$
$\displaystyle = \int \big(\frac{3}{8}+\frac{1}{2}\cos 2ax+\frac{1}{8}\cos 4ax\big)\,\mathrm{d}x = \frac{3}{8}x+\frac{1}{4a}\sin 2ax+\frac{1}{32a}\sin 4ax + c$

習題. 以變數變換法求下列不定積分. 注意: 可能會因為常數項而跟此處答案不同.

1. $\displaystyle\int \frac{1}{\sqrt{2x-1}}\,\mathrm{d}x = \sqrt{2x-1}+c$
2. $\displaystyle\int \sqrt{7x+4}\,\mathrm{d}x = \frac{2}{21}\,(7x+4)^{\frac{3}{2}}+c$
3. $\displaystyle\int \sin(3x-1)\,\mathrm{d}x = \frac{-\cos(3x-1)}{3}+c$
4. $\displaystyle\int e^{\pi x-1}\,\mathrm{d}x = \frac{e^{\pi x-1}}{\pi}+c$
5. $\displaystyle\int (x^2-2x+1)^{\frac{1}{3}}\,\mathrm{d}x = \frac{3}{5}\,(x-1)^{\frac{5}{3}}+c$
6. $\displaystyle\int \sin 3x\cos 3x\,\mathrm{d}x = \frac{\sin^2 3x}{6}+c$
7. $\displaystyle\int \frac{\cos 3x}{\sin^2 3x}\,\mathrm{d}x = \frac{-1}{3\sin 3x}+c$
8. $\displaystyle\int \frac{x}{\sqrt{1+2x^2}}\,\mathrm{d}x = \frac{\sqrt{1+2x^2}}{2}+c$
9. $\displaystyle\int x^2\sqrt{1-x}\,\mathrm{d}x = -\frac{2\,(1-x)^{\frac{3}{2}}}{3}+\frac{4\,(1-x)^{\frac{5}{2}}}{5}-\frac{2\,(1-x)^{\frac{7}{2}}}{7}$

解.

1. 令 $u=2x-1$, 則 $\mathrm{d}u=2\,\mathrm{d}x \implies \mathrm{d}x=\frac{1}{2}\,\mathrm{d}u$. 故 $\displaystyle\int \frac{1}{\sqrt{2x-1}}\,\mathrm{d}x = \int \frac{1}{\sqrt{u}}\cdot\frac{1}{2}\,\mathrm{d}u = \sqrt{u}+c = \sqrt{2x-1}+c.$
2. 令 $u=7x+4$, 則 $\mathrm{d}u=7\,\mathrm{d}x \implies \mathrm{d}x=\frac{1}{7}\,\mathrm{d}u$. 故 $\displaystyle\int \sqrt{7x+4}\,\mathrm{d}x = \int \sqrt{u}\cdot\frac{1}{7}\,\mathrm{d}u = \frac{1}{7}\cdot\frac{2}{3}\,u^{\frac{3}{2}}+c = \frac{2}{21}\,(7x+4)^{\frac{3}{2}}+c.$
3. 令 $u=3x-1$, 則 $\mathrm{d}u=3\,\mathrm{d}x \implies \mathrm{d}x=\frac{1}{3}\,\mathrm{d}u$. 故 $\displaystyle\int \sin(3x-1)\,\mathrm{d}x = \int \sin u\cdot\frac{1}{3}\,\mathrm{d}u = -\frac{1}{3}\cos u+c = \frac{-\cos(3x-1)}{3}+c.$
4. 令 $u=\pi x-1$, 則 $\mathrm{d}u=\pi\,\mathrm{d}x \implies \mathrm{d}x=\frac{1}{\pi}\,\mathrm{d}u$. 故 $\displaystyle\int e^{\pi x-1}\,\mathrm{d}x = \int e^u\cdot\frac{1}{\pi}\,\mathrm{d}u = \frac{1}{\pi}\,e^u+c = \frac{e^{\pi x-1}}{\pi}+c.$
5. 令 $u=x-1$, 則 $\mathrm{d}u=\mathrm{d}x$. 故 $\displaystyle\int (x^2-2x+1)^{\frac{1}{3}}\,\mathrm{d}x = \int (x-1)^{\frac{2}{3}}\,\mathrm{d}x = \int u^{\frac{2}{3}}\,\mathrm{d}u = \frac{3}{5}\,u^{\frac{5}{3}}+c = \frac{3}{5}\,(x-1)^{\frac{5}{3}}+c.$
6. 令 $u=\sin 3x$, 則 $\mathrm{d}u=3\cos 3x\,\mathrm{d}x \implies \cos 3x\,\mathrm{d}x=\frac{1}{3}\,\mathrm{d}u$. 故 $\displaystyle\int \sin 3x\cos 3x\,\mathrm{d}x = \int u\cdot\frac{1}{3}\,\mathrm{d}u = \frac{1}{3}\cdot\frac{1}{2}\,u^2+c = \frac{\sin^2 3x}{6}+c.$
7. 令 $u=\sin 3x$, 則 $\mathrm{d}u=3\cos 3x\,\mathrm{d}x \implies \cos 3x\,\mathrm{d}x=\frac{1}{3}\,\mathrm{d}u$. 故 $\displaystyle\int \frac{\cos 3x}{\sin^2 3x}\,\mathrm{d}x = \int \frac{1}{u^2}\cdot\frac{1}{3}\,\mathrm{d}u = -\frac{1}{3}\cdot\frac{1}{u}+c = -\frac{1}{3\sin 3x}+c.$
8. 令 $u=1+2x^2$, 則 $\mathrm{d}u=4x\,\mathrm{d}x \implies x\,\mathrm{d}x=\frac{1}{4}\,\mathrm{d}u$. 故 $\displaystyle\int \frac{x}{\sqrt{1+2x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{1+2x^2}}\cdot x\,\mathrm{d}x = \int \frac{1}{\sqrt{u}}\cdot\frac{1}{4}\,\mathrm{d}u = \frac{1}{4}\cdot 2\,u^{\frac{1}{2}}+c = \frac{\sqrt{1+2x^2}}{2}+c.$
9. 令 $u=\sqrt{1-x}$, 則 $u^2=1-x \implies x=1-u^2$, $\mathrm{d}x=-2\,u\,\mathrm{d}u$. 故 $\displaystyle\int x^2\sqrt{1-x}\,\mathrm{d}x = \int (1-u^2)^2\cdot u\cdot(-2)\,u\,\mathrm{d}u = -2\int (1-u^2)^2\cdot u^2\,\mathrm{d}u = -2\int \big(u^2-2u^4+u^6\big)\,\mathrm{d}u = -\frac{2\,u^3}{3}+\frac{4\,u^5}{5}-\frac{2\,u^7}{7}+c = -\frac{2\,(1-x)^{\frac{3}{2}}}{3}+\frac{4\,(1-x)^{\frac{5}{2}}}{5}-\frac{2\,(1-x)^{\frac{7}{2}}}{7}.$

習題. 以變數變換法求下列不定積分, 其中 $a \neq b \neq 0$. 注意: 可能會因為常數項而跟此處答案不同.

1. $\displaystyle\int e^{2x}\sin e^{2x}\,\mathrm{d}x = -\frac{1}{2}\cos e^{2x} + c$
2. $\displaystyle\int xe^{-\frac{x^2}{2}}\,\mathrm{d}x = -e^{-\frac{x^2}{2}} + c$
3. $\displaystyle\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,\mathrm{d}x = -2\cos\sqrt{x} + c$
4. $\displaystyle\int x^2 2^{x^3+1}\,\mathrm{d}x = \frac{2^{x^3+1}}{3\ln 2} + c$
5. $\displaystyle\int \frac{\ln x}{x}\,\mathrm{d}x = \frac{1}{2}(\ln x)^2 + c$
6. $\displaystyle\int \frac{x+1}{\sqrt{x^2+2x+3}}\,\mathrm{d}x = \sqrt{x^2+2x+3} + c$
7. $\displaystyle\int \frac{x^2}{2+x^6}\,\mathrm{d}x = \frac{1}{3\sqrt{2}}\tan^{-1}\frac{x^3}{\sqrt{2}} + c$
8. $\displaystyle\int \frac{1}{e^x+e^{-x}}\,\mathrm{d}x = \tan^{-1} e^x + c$
9. $\displaystyle\int \frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x = -\sqrt{1-x^2} + \sin^{-1} x + c$
10. $\displaystyle\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,\mathrm{d}x = \sin^{-1}(\tan x) + c$
11. $\displaystyle\int \sin^4 x\cos^5 x\,\mathrm{d}x = \frac{\sin^5 x}{5} - \frac{2\sin^7 x}{7} + \frac{\sin^9 x}{9}$
12. $\displaystyle\int \sin^2 x\cos^2 x\,\mathrm{d}x = \frac{x}{8} - \frac{\sin 4x}{32} + c$
13. $\displaystyle\int \sin^{-\frac{2}{3}} x\cos^3 x\,\mathrm{d}x = 3\sin^{\frac{1}{3}} x - \frac{3}{7}\sin^{\frac{7}{3}} x + c$
14. $\displaystyle\int \frac{\sin^3(\ln x)\cos^3(\ln x)}{x}\,\mathrm{d}x = \frac{\sin^4(\ln x)}{4} - \frac{\sin^6(\ln x)}{6} + c$
15. $\displaystyle\int \frac{\sin^3 x}{\cos^4 x}\,\mathrm{d}x = -\frac{1}{\cos x} + \frac{1}{3\cos^3 x} + c$
16. $\displaystyle\int \cos ax\cos bx\,\mathrm{d}x = \frac{\sin(a-b)x}{2(a-b)} + \frac{\sin(a+b)x}{2(a+b)} + c$
17. $\displaystyle\int \sin ax\sin bx\,\mathrm{d}x = \frac{\sin(a-b)x}{2(a-b)} - \frac{\sin(a+b)x}{2(a+b)} + c$
18. $\displaystyle\int \sin ax\cos bx\,\mathrm{d}x = -\frac{\cos(a-b)x}{2(a-b)} - \frac{\cos(a+b)x}{2(a+b)} + c$

解.

1. 令 $u = e^{2x}$, 則 $\mathrm{d}u = 2e^{2x}\,\mathrm{d}x \implies e^{2x}\,\mathrm{d}x = \frac{1}{2}\mathrm{d}u$. 故 $\displaystyle\int e^{2x}\sin e^{2x}\,\mathrm{d}x = \int \sin e^{2x}\cdot e^{2x}\,\mathrm{d}x = \int \sin u\cdot\frac{1}{2}\,\mathrm{d}u = -\frac{1}{2}\cos u + c = -\frac{1}{2}\cos e^{2x} + c$.
2. 令 $u = \frac{x^2}{2}$, 則 $\mathrm{d}u = x\,\mathrm{d}x$, 故 $\displaystyle\int xe^{-\frac{x^2}{2}}\,\mathrm{d}x = \int e^{-\frac{x^2}{2}}\cdot x\,\mathrm{d}x = \int e^{-u}\,\mathrm{d}u = -e^{-u} + c = -e^{-\frac{x^2}{2}} + c$
3. 令 $u = \sqrt{x}$, 則 $\mathrm{d}u = \frac{1}{2\sqrt{x}}\,\mathrm{d}x \implies \frac{1}{\sqrt{x}}\,\mathrm{d}x = 2\,\mathrm{d}u$, 故 $\displaystyle\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,\mathrm{d}x = \int \sin\sqrt{x}\cdot\frac{1}{\sqrt{x}}\,\mathrm{d}x = \int \sin u\cdot 2\,\mathrm{d}u = -2\cos u + c = -2\cos\sqrt{x} + c$
4. 令 $u = x^3+1$, 則 $\mathrm{d}u = 3x^2\,\mathrm{d}x \implies x^2\,\mathrm{d}x = \frac{1}{3}\mathrm{d}u$, 故 $\displaystyle\int x^2 2^{x^3+1}\,\mathrm{d}x = \int 2^{x^3+1}\cdot x^2\,\mathrm{d}x = \int 2^u\cdot\frac{1}{3}\,\mathrm{d}u = \frac{1}{3}\int e^{u\ln 2}\,\mathrm{d}u = \frac{1}{3\ln 2}e^{u\ln 2} + c = \frac{2^{x^3+1}}{3\ln 2} + c$
5. 令 $u = \ln x$, 則 $\mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$, 故 $\displaystyle\int \frac{\ln x}{x}\,\mathrm{d}x = \int \ln x\cdot\frac{1}{x}\,\mathrm{d}x = \int u\,\mathrm{d}u = \frac{1}{2}u^2 + c = \frac{1}{2}(\ln x)^2 + c$
6. 令 $u = x^2+2x+3$, 則 $\mathrm{d}u = (2x+2)\,\mathrm{d}x \implies (x+1)\,\mathrm{d}x = \frac{1}{2}\mathrm{d}u$, 故 $\displaystyle\int \frac{x+1}{\sqrt{x^2+2x+3}}\,\mathrm{d}x = \int \frac{1}{\sqrt{x^2+2x+3}}\cdot(x+1)\,\mathrm{d}x = \int \frac{1}{\sqrt{u}}\cdot\frac{1}{2}\,\mathrm{d}u = \sqrt{u} + c = \sqrt{x^2+2x+3} + c$
7. 令 $u = x^3$, 則 $\mathrm{d}u = 3x^2\,\mathrm{d}x \implies x^2\,\mathrm{d}x = \frac{1}{3}\mathrm{d}u$, 故 $\displaystyle\int \frac{x^2}{2+x^6}\,\mathrm{d}x = \int \frac{1}{2+x^6}\cdot x^2\,\mathrm{d}x = \int \frac{1}{2+u^2}\cdot\frac{1}{3}\,\mathrm{d}u = \frac{1}{3\sqrt{2}}\tan^{-1}\frac{u}{\sqrt{2}} + c = \frac{1}{3\sqrt{2}}\tan^{-1}\frac{x^3}{\sqrt{2}} + c$

8. $\int \frac{1}{e^x+e^{-x}}\,\mathrm{d}x = \int \frac{e^x}{e^{2x}+1}\,\mathrm{d}x$. 令 $u = e^x$, 則 $\mathrm{d}u = e^x\,\mathrm{d}x$, 故 $\int \frac{e^x}{e^{2x}+1}\,\mathrm{d}x = \int \frac{1}{e^{2x}+1}\cdot e^x\,\mathrm{d}x = \int \frac{1}{u^2+1}\,\mathrm{d}u = \tan^{-1}u + c = \tan^{-1}e^x + c$.

9. $\int \frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x = \int \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x + \int \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x$. 令 $u = \sqrt{1-x^2}$, 則 $\mathrm{d}u = \frac{-x}{\sqrt{1-x^2}}\,\mathrm{d}x \implies \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x = -\mathrm{d}u$, 故 $\int \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x = \int -\mathrm{d}u = -u + c = -\sqrt{1-x^2}$, $\int \frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x = \int \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x + \int \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x = -\sqrt{1-x^2} + \sin^{-1}x + c$.

10. 令 $u = \tan x$, 則 $\mathrm{d}u = \sec^2 x\,\mathrm{d}x$, 故 $\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,\mathrm{d}x = \int \frac{1}{\sqrt{1-\tan^2 x}}\cdot\sec^2 x\,\mathrm{d}x = \int \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u = \sin^{-1}u + c = \sin^{-1}(\tan x) + c$

11. $\int \sin^4 x\cos^5 x\,\mathrm{d}x = \int \sin^4 x\cdot\cos^4 x\cdot\cos x\,\mathrm{d}x = \int \sin^4 x\cdot(1-\sin^2 x)^2\cdot\cos x\,\mathrm{d}x = \int \big(\sin^4 x - 2\sin^6 x + \sin^8 x\big)\cdot\cos x\,\mathrm{d}x$. 令 $u = \sin x$, 則 $\mathrm{d}u = \cos x\,\mathrm{d}x$, 故 $\int \big(\sin^4 x - 2\sin^6 x + \sin^8 x\big)\cdot\cos x\,\mathrm{d}x = \int \big(u^4 - 2u^6 + u^8\big)\,\mathrm{d}u = \frac{u^5}{5} - \frac{2u^7}{7} + \frac{u^9}{9} + c = \frac{\sin^5 x}{5} - \frac{2\sin^7 x}{7} + \frac{\sin^9 x}{9} + c$

12. $\int \sin^2 x\cos^2 x\,\mathrm{d}x = \int \frac{1-\cos 2x}{2}\cdot\frac{1+\cos 2x}{2}\,\mathrm{d}x = \frac{1}{4}\int \big(1-\cos^2 2x\big)\,\mathrm{d}x = \frac{1}{4}\int \big(1 - \frac{1+\cos 4x}{2}\big)\,\mathrm{d}x = \frac{1}{4}\int \big(\frac{1}{2} + \frac{\cos 4x}{2}\big)\,\mathrm{d}x = \frac{x}{8} - \frac{\sin 4x}{32} + c$

13. $\int \sin^{-\frac{2}{3}} x\cos^3 x\,\mathrm{d}x = \int \sin^{-\frac{2}{3}} x\cdot\cos^2 x\cdot\cos x\,\mathrm{d}x = \int \sin^{-\frac{2}{3}} x\cdot(1-\sin^2 x)\cdot\cos x\,\mathrm{d}x = \int \big(\sin^{-\frac{2}{3}} x - \sin^{\frac{4}{3}} x\big)\cdot\cos x\,\mathrm{d}x$. 令 $u = \sin x$, 則 $\mathrm{d}u = \cos x\,\mathrm{d}x$, 故 $\int \big(\sin^{-\frac{2}{3}} x - \sin^{\frac{4}{3}} x\big)\cdot\cos x\,\mathrm{d}x = \int \big(u^{-\frac{2}{3}} - u^{\frac{4}{3}}\big)\,\mathrm{d}u = 3u^{\frac{1}{3}} - \frac{3}{7}u^{\frac{7}{3}} + c = 3\sin^{\frac{1}{3}} x - \frac{3}{7}\sin^{\frac{7}{3}} x + c$

14. 令 $u = \ln x$, 則 $\mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$, 故 $\int \frac{\sin^3(\ln x)\cos^3(\ln x)}{x}\,\mathrm{d}x = \int \sin^3(\ln x)\cos^3(\ln x)\cdot\frac{1}{x}\,\mathrm{d}x = \int \sin^3 u\cos^3 u\,\mathrm{d}u = \int \sin^3 u\cdot\cos^2 u\cdot\cos u\,\mathrm{d}u = \int \sin^3 u\big(1-\sin^2 u\big)\cdot\cos u\,\mathrm{d}u = \int \big(\sin^3 u - \sin^5 u\big)\cdot\cos u\,\mathrm{d}u$. 令 $w = \sin u$, 則 $\mathrm{d}w = \cos u\,\mathrm{d}u$, 故 $\int \big(\sin^3 u - \sin^5 u\big)\cdot\cos u\,\mathrm{d}u = \int \big(w^3 - w^5\big)\,\mathrm{d}w = \frac{w^4}{4} - \frac{w^6}{6} + c = \frac{\sin^4(\ln x)}{4} - \frac{\sin^6(\ln x)}{6} + c$

15. $\int \frac{\sin^3 x}{\cos^4 x}\,\mathrm{d}x = \int \frac{\sin^2 x\cdot\sin x}{\cos^4 x}\,\mathrm{d}x = \int \frac{\sin^2 x}{\cos^4 x}\cdot\sin x\,\mathrm{d}x = \int \frac{1-\cos^2 x}{\cos^4 x}\cdot\sin x\,\mathrm{d}x = \int \big(\frac{1}{\cos^4 x} - \frac{1}{\cos^2 x}\big)\cdot\sin x\,\mathrm{d}x$. 令 $u = \cos x$, 則 $\mathrm{d}u = -\sin x\,\mathrm{d}x \implies \sin x\,\mathrm{d}x = -\mathrm{d}u$, 故 $\int \big(\frac{1}{\cos^4 x} - \frac{1}{\cos^2 x}\big)\cdot\sin x\,\mathrm{d}x = \int \big(\frac{1}{u^4} - \frac{1}{u^2}\big)(-)\,\mathrm{d}u = \int \big(\frac{1}{u^2} - \frac{1}{u^4}\big)\,\mathrm{d}u = -\frac{1}{u} + \frac{1}{3u^3} + c = -\frac{1}{\cos x} + \frac{1}{3\cos^3 x} + c$

16. 由積化和差公式 $\cos\alpha\cos\beta = \frac{1}{2}\big(\cos(\alpha-\beta) + \cos(\alpha+\beta)\big)$, $\int \cos ax\cos bx\,\mathrm{d}x = \int \frac{1}{2}\big(\cos(ax-bx) + \cos(ax+bx)\big)\,\mathrm{d}x = \int \frac{1}{2}\big(\cos(a-b)x + \cos(a+b)x\big)\,\mathrm{d}x = \frac{\sin(a-b)x}{2(a-b)} + \frac{\sin(a+b)x}{2(a+b)} + c$.

17. 由積化和差公式 $\sin\alpha\sin\beta = \frac{1}{2}\big(\cos(\alpha-\beta) - \cos(\alpha+\beta)\big)$, $\int \sin ax\sin bx\,\mathrm{d}x = \int \frac{1}{2}\big(\cos(ax-bx) - \cos(ax+bx)\big)\,\mathrm{d}x = \int \frac{1}{2}\big(\cos(a-b)x - \cos(a+b)x\big)\,\mathrm{d}x = \frac{\sin(a-b)x}{2(a-b)} - \frac{\sin(a+b)x}{2(a+b)} + c$.

18. 由積化和差公式 $\sin\alpha\cos\beta = \frac{1}{2}\big(\sin(\alpha-\beta)+\sin(\alpha+\beta)\big)$, $\int \sin ax\cos bx\,\mathrm{d}x = \int \frac{1}{2}\big(\sin(ax-bx)+\sin(ax+bx)\big)\,\mathrm{d}x = \int \frac{1}{2}\big(\sin(a-b)x+\sin(a+b)x\big)\,\mathrm{d}x = -\frac{\cos(a-b)x}{2(a-b)} - \frac{\cos(a+b)x}{2(a+b)} + c.$

部份積分法

結論. $\frac{\mathrm{d}}{\mathrm{d}x}\left(u(x)\,v(x)\right) = u(x)\,\frac{\mathrm{d}v(x)}{\mathrm{d}x} + v(x)\,\frac{\mathrm{d}u(x)}{\mathrm{d}x} \implies u(x)\,v(x) = \int u(x)\,\mathrm{d}v(x) + \int v(x)\,\mathrm{d}u(x)$
$\implies \int u\,\mathrm{d}v = u\,v - \int v\,\mathrm{d}u.$

例. 求 $\int x\,e^x\,\mathrm{d}x.$

解. 令 $u = x$, 則 $\mathrm{d}u = \mathrm{d}x$. 令 $\mathrm{d}v = e^x\,\mathrm{d}x$, 則 $v = e^x$. 故 $\int x\,e^x\,\mathrm{d}x = x\cdot e^x - \int e^x\,\mathrm{d}x = xe^x - e^x + c.$

例. 求 $\int \ln x\,\mathrm{d}x.$

解. 令 $u = \ln x$, 則 $\mathrm{d}u = \frac{1}{x}\mathrm{d}x$. 令 $\mathrm{d}v = \mathrm{d}x$, 則 $v = x$. 故 $\int \ln x\,\mathrm{d}x = \ln x\cdot x - \int x\cdot\frac{1}{x}\,\mathrm{d}x = x\ln x - x + c.$

例. 求 $\int x\tan^{-1}x\,\mathrm{d}x.$

解. 令 $u = \tan^{-1}x$, 則 $\mathrm{d}u = \frac{1}{1+x^2}\mathrm{d}x$. 令 $\mathrm{d}v = x\,\mathrm{d}x$, 則 $v = \frac{x^2}{2}$. 故 $\int x\tan^{-1}x\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^2}{2} - \frac{1}{2}\int\frac{x^2}{1+x^2}\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^2}{2} - \frac{1}{2}\int\big(1-\frac{1}{1+x^2}\big)\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^2}{2} - \frac{1}{2}\big(x-\tan^{-1}x\big) + c.$

例. 求 $\int x^2\sin x\,\mathrm{d}x.$

解. 令 $u = x^2$, 則 $\mathrm{d}u = 2x\,\mathrm{d}x$; 令 $\mathrm{d}v = \sin x\,\mathrm{d}x$, 則 $v = -\cos x$. 故 $\int x^2\,\sin x\,\mathrm{d}x = x^2\cdot(-\cos x) + 2\int x\cos x\,\mathrm{d}x.$
令 $u = x$, 則 $\mathrm{d}u = \mathrm{d}x$; 令 $\mathrm{d}v = \cos x\,\mathrm{d}x$, 則 $v = \sin x$. 故 $\int x\cos x\,\mathrm{d}x = x\cdot\sin x - \int\sin x\,\mathrm{d}x = x\cdot\sin x + \cos x.$
由上, $\int x^2\sin x\,\mathrm{d}x = -x^2\cos x + 2(x\sin x + \cos x) + c.$

例. 求 $\int \sin^{-1}\sqrt{1-x^2}\,\mathrm{d}x$, $x > 0$.

解. 令 $u = \sin^{-1}\sqrt{1-x^2}$, 則 $\mathrm{d}u = \frac{1}{\sqrt{1-\left(\sqrt{1-x^2}\right)^2}}\cdot\frac{-x}{\sqrt{1-x^2}}\,\mathrm{d}x = \frac{1}{\sqrt{x^2}}\cdot\frac{-x}{\sqrt{1-x^2}}\,\mathrm{d}x = \frac{-1}{\sqrt{1-x^2}}\,\mathrm{d}x$; 令 $\mathrm{d}v = \mathrm{d}x$, 則 $v = x$. 故 $\int \sin^{-1}\sqrt{1-x^2}\,\mathrm{d}x = x\,\sin^{-1}\sqrt{1-x^2} + \int\frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x = x\,\sin^{-1}\sqrt{1-x^2} - \sqrt{1-x^2}.$

例. 求 $\int e^{ax}\cos bx\,\mathrm{d}x$ 與 $\int e^{ax}\sin bx\,\mathrm{d}x$, a, $b \neq 0$.

解.

(a) 考慮 $\int e^{ax}\cos bx\,\mathrm{d}x$: 令 $u=\cos bx$, 則 $\mathrm{d}u=-b\sin bx\,\mathrm{d}x$; 令 $\mathrm{d}v=e^{ax}\,\mathrm{d}x$, 則 $v=\dfrac{1}{a}\,e^{ax}$, 故 $\int e^{ax}\cos bx\,\mathrm{d}x=\cos bx\cdot\dfrac{1}{a}e^{ax}+\dfrac{b}{a}\int e^{ax}\cdot\sin bx\,\mathrm{d}x$.

(b) 考慮 $\int e^{ax}\sin bx\,\mathrm{d}x$: 令 $u=\sin bx$, 則 $\mathrm{d}u=b\cos bx\,\mathrm{d}x$; 令 $\mathrm{d}v=e^{ax}\,\mathrm{d}x$, 則 $v=\dfrac{1}{a}\,e^{ax}$, 故 $\int e^{ax}\sin bx\,\mathrm{d}x=\sin bx\cdot\dfrac{1}{a}\,e^{ax}-\dfrac{b}{a}\int e^{ax}\cdot\cos bx\,\mathrm{d}x$.

令 $X=\int e^{ax}\cos bx\,\mathrm{d}x$, $Y=\int e^{ax}\sin bx\,\mathrm{d}x$, 由 (a)(b) $X=\cos bx\cdot\dfrac{1}{a}e^{ax}+\dfrac{b}{a}Y$, $Y=\sin bx\cdot\dfrac{1}{a}e^{ax}-\dfrac{b}{a}X$. 解 X, Y 得 $X=\dfrac{e^{ax}(a\cos bx+b\sin bx)}{a^2+b^2}$, $Y=\dfrac{e^{ax}(a\sin bx-b\cos bx)}{a^2+b^2}$.

列表法

結論.

1. 將要微分函數寫左邊, 積分函數寫右邊; 左邊連續微分, 右邊連續積分
2. 依序左上連右下斜線函數相乘, 最底部水平兩邊函數相乘並積分, 符號正負相間
3. 將上式所得項全部加總即為所求積分

例. 求 $\int (x+3)\,e^{2x}\,\mathrm{d}x$.

解.

Diff		Int
$x+3$	$+$	e^{2x}
1	$-$	$\frac{1}{2}e^{2x}$
0		$\frac{1}{4}e^{2x}$

$$\int (x+3)\,e^{2x}\,\mathrm{d}x=(x+3)\,\frac{1}{2}e^{2x}-\frac{1}{4}e^{2x}$$

例. 求 $\int (x^2-2x)\,e^{kx}\,\mathrm{d}x$.

解.

Diff		Int
x^2-2x	$+$	e^{kx}
$2(x-1)$	$-$	$\frac{1}{k}e^{kx}$
2	$+$	$\frac{1}{k^2}e^{kx}$
0		$\frac{1}{k^3}e^{kx}$

$$\int (x^2-2x)\,e^{kx}\,\mathrm{d}x=(x^2-2x)\,\frac{1}{k}e^{kx}-2(x-1)\frac{1}{k^2}e^{kx}+2\frac{1}{k^3}e^{kx}$$

例. 求 $\int x^4\,\sin 2x\,\mathrm{d}x$.

解.

Diff		Int
x^4	$+$	$\sin 2x$
$4x^3$	$-$	$-\frac{1}{2}\cos 2x$
$12x^2$	$+$	$-\frac{1}{4}\sin 2x$
$24x$	$-$	$\frac{1}{8}\cos 2x$
24	$+$	$\frac{1}{16}\sin 2x$
0		$-\frac{1}{32}\cos 2x$

$$\begin{aligned}&\int x^4\,\sin 2x\,\mathrm{d}x\\&=-x^4\,\frac{1}{2}\cos 2x+4x^3\,\frac{1}{4}\sin 2x+12x^2\,\frac{1}{8}\cos 2x-24x\,\frac{1}{16}\sin 2x-24\,\frac{1}{32}\cos 2x\\&=\left(-\frac{x^4}{2}+\frac{3x^2}{2}-\frac{3}{4}\right)\cos 2x+\left(x^3-\frac{3x}{2}\right)\sin 2x\end{aligned}$$

例. 求 $\displaystyle\int x^5\, e^{ax}\,\mathrm{d}x$.

解.

Diff		Int
x^5	$+$	e^{ax}
$5x^4$	$-$	$\frac{1}{a}e^{ax}$
$20x^3$	$+$	$\frac{1}{a^2}e^{ax}$
$60x^2$	$-$	$\frac{1}{a^3}e^{ax}$
$120x$	$+$	$\frac{1}{a^4}e^{ax}$
120	$-$	$\frac{1}{a^5}e^{ax}$
0		$\frac{1}{a^6}e^{ax}$

$$\begin{aligned}
&\int x^5\, e^{ax}\,\mathrm{d}x \\
&= x^5\,\frac{1}{a}e^{ax} - 5x^4\,\frac{1}{a^2}e^{ax} + 20x^3\,\frac{1}{a^3}e^{ax} - 60x^2\,\frac{1}{a^4}e^{ax} + 120x\,\frac{1}{a^5}e^{ax} - 120\,\frac{1}{a^6}e^{ax} \\
&= \left(\frac{x^5}{a} - \frac{5x^4}{a^2} + \frac{20x^3}{a^3} - \frac{60x^2}{a^4} + \frac{120x}{a^5} - \frac{120}{a^6}\right) e^{ax}
\end{aligned}$$

例. 求 $\displaystyle\int (\sin^{-1} x)^2\,\mathrm{d}x$.

解.

Diff		Int
$(\sin^{-1} x)^2$	$+$	1
$\frac{2\sin^{-1} x}{\sqrt{1-x^2}}$	$-$	x

Diff		Int
$2\sin^{-1} x$	$+$	$\frac{x}{\sqrt{1-x^2}}$
$\frac{2}{\sqrt{1-x^2}}$	$-$	$-\sqrt{1-x^2}$

$$\begin{aligned}
&\int (\sin^{-1} x)^2\,\mathrm{d}x = (\sin^{-1} x)^2 \cdot x - \int \frac{2\sin^{-1} x}{\sqrt{1-x^2}} \cdot x\,\mathrm{d}x \\
&= (\sin^{-1} x)^2 \cdot x - \int 2\sin^{-1} x \cdot \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x \\
&= (\sin^{-1} x)^2 \cdot x - \left(-2\sin^{-1} x \cdot \sqrt{1-x^2} + \int \frac{2}{\sqrt{1-x^2}} \cdot \sqrt{1-x^2}\,\mathrm{d}x\right) \\
&= (\sin^{-1} x)^2 \cdot x + 2\sin^{-1} x \cdot \sqrt{1-x^2} - 2x
\end{aligned}$$

例. 求 $\displaystyle\int e^{ax}\cos bx\,\mathrm{d}x$, $a,\ b \neq 0$.

解.

Diff		Int
$\cos bx$	$+$	e^{ax}
$-b\sin bx$	$-$	$\frac{1}{a}e^{ax}$
$-b^2\cos bx$	$+$	$\frac{1}{a^2}e^{ax}$

$$\begin{aligned}
&\int e^{ax}\cos bx\,\mathrm{d}x = \frac{1}{a}e^{ax}\cos bx + \frac{b}{a^2}e^{ax}\sin bx - \frac{b^2}{a^2}\int e^{ax}\cos bx\,\mathrm{d}x \\
&\Longrightarrow \int e^{ax}\cos bx\,\mathrm{d}x = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}
\end{aligned}$$

例. 求 $\displaystyle\int e^{ax}\sin bx\,\mathrm{d}x$, $a,\ b \neq 0$.

解.

Diff		Int
$\sin bx$	$+$	e^{ax}
$b\cos bx$	$-$	$\frac{1}{a}e^{ax}$
$-b^2\sin bx$	$+$	$\frac{1}{a^2}e^{ax}$

$$\begin{aligned}
&\int e^{ax}\sin bx\,\mathrm{d}x = \frac{1}{a}e^{ax}\sin bx - \frac{b}{a^2}e^{ax}\cos bx - \frac{b^2}{a^2}\int e^{ax}\sin bx\,\mathrm{d}x \\
&\Longrightarrow \int e^{ax}\sin bx\,\mathrm{d}x = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}
\end{aligned}$$

遞迴式

例. 求 $I_n = \displaystyle\int \frac{1}{(x^2+a^2)^n}\,\mathrm{d}x$, $n \in \mathbb{N}$ 之遞迴式: 令 $u = \dfrac{1}{(x^2+a^2)^n}$, 則 $\mathrm{d}u = -2n\,\dfrac{x}{(x^2+a^2)^{n+1}}\,\mathrm{d}x$; $\mathrm{d}v = \mathrm{d}x$, 則 $v = x$. 故 $I_n = \displaystyle\int \frac{1}{(x^2+a^2)^n}\,\mathrm{d}x = \frac{x}{(x^2+a^2)^n} + 2n\int \frac{x^2}{(x^2+a^2)^{n+1}}\,\mathrm{d}x = \frac{x}{(x^2+a^2)^n} + 2n\int \frac{(x^2+a^2-a^2)}{(x^2+a^2)^{n+1}}\,\mathrm{d}x =$

$\dfrac{x}{(x^2+a^2)^n}+2nI_n-2na^2I_{n+1} \implies 2na^2I_{n+1}=\dfrac{x}{(x^2+a^2)^n}+(2n-1)I_n \implies I_{n+1}=\dfrac{x}{2na^2(x^2+a^2)^n}+\dfrac{2n-1}{2na^2}I_n.$

使用例: $I_1=\dfrac{1}{a}\tan^{-1}\dfrac{x}{a}$, $I_2=\dfrac{1}{2a^2}I_1+\dfrac{x}{2a^2(x^2+a^2)}=\dfrac{1}{2a^3}\tan^{-1}\dfrac{x}{a}+\dfrac{x}{2a^2(x^2+a^2)}$, $I_3=\dfrac{3}{4a^2}I_2+\dfrac{x}{4a^2(x^2+a^2)^2}$ $=\dfrac{3}{4a^2}\left(\dfrac{1}{2a^3}\tan^{-1}\dfrac{x}{a}+\dfrac{x}{2a^2(x^2+a^2)}\right)+\dfrac{x}{4a^2(x^2+a^2)^2}=\dfrac{3}{8a^5}\tan^{-1}\dfrac{x}{a}+\dfrac{3x}{8a^4(x^2+a^2)}+\dfrac{x}{4a^2(x^2+a^2)^2}.$

例. 求 $J_n=\displaystyle\int\sin^n x\,\mathrm{d}x$, $n\in\mathbb{N}$, $n\geqslant 2$ 之遞迴式: 令 $u=\sin^{n-1}x$, 則 $\mathrm{d}u=(n-1)\sin^{n-2}x\cos x\,\mathrm{d}x$; $\mathrm{d}v=\sin x\,\mathrm{d}x$, 則 $v=-\cos x$. 故 $J_n=\displaystyle\int\sin^n x\,\mathrm{d}x=-\sin^{n-1}x\cdot\cos x+(n-1)\int\cos x\cdot\sin^{n-2}x\cos x\,\mathrm{d}x=-\sin^{n-1}x\cos x+(n-1)\int\sin^{n-2}x\cos^2 x\,\mathrm{d}x=-\sin^{n-1}x\cos x+(n-1)\int\sin^{n-2}x\cdot(1-\sin^2 x)\,\mathrm{d}x=-\sin^{n-1}x\cos x+(n-1)\int\sin^{n-2}x\,\mathrm{d}x-(n-1)\int\sin^n x\,\mathrm{d}x=-\sin^{n-1}x\cos x+(n-1)J_{n-2}+(1-n)J_n \implies nJ_n=-\sin^{n-1}x\cos x+(n-1)J_{n-2} \implies J_n=\dfrac{-\sin^{n-1}x\cos x}{n}+\dfrac{n-1}{n}J_{n-2}.$

使用例: $J_2=\dfrac{-\sin x\cos x}{2}+\dfrac{1}{2}J_0=\dfrac{-\sin x\cos x}{2}+\dfrac{x}{2}$, $J_3=\dfrac{-\sin^2 x\cos x}{3}+\dfrac{2}{3}J_1=\dfrac{-\sin^2 x\cos x}{3}-\dfrac{2\cos x}{3}$, $J_4=\dfrac{-\sin^3 x\cos x}{4}+\dfrac{3}{4}J_2=\dfrac{-\sin^3 x\cos x}{4}+\dfrac{3\sin x\cos x}{8}-\dfrac{3x}{8}.$

習題. 以部份積分法求下列不定積分. 注意: 可能會因為常數項而跟此處答案不同.

1. $\displaystyle\int x^3\ln x\,\mathrm{d}x=\frac{x^4}{4}\ln x-\frac{x^4}{16}+c$
2. $\displaystyle\int x(\ln x)^3\,\mathrm{d}x=\frac{x^2}{2}\left((\ln x)^3-\frac{3(\ln x)^2}{2}+\frac{3\ln x}{2}-\frac{3}{4}\right)+c$
3. $\displaystyle\int x^2\tan^{-1}x\,\mathrm{d}x=\frac{x^3}{3}\tan^{-1}x-\frac{x^2}{6}+\frac{\ln(1+x^2)}{6}+c$
4. $\displaystyle\int x^5e^{-x^2}\,\mathrm{d}x=-\frac{1}{2}e^{-x^2}(x^4+2x^2+2)+c$
5. $\displaystyle\int xe^{\sqrt{x}}\,\mathrm{d}x=2\,e^{\sqrt{x}}\left(x\sqrt{x}-3x+6\sqrt{x}-6\right)+c$
6. $\displaystyle\int\frac{xe^x}{(x+1)^2}\,\mathrm{d}x=\frac{e^x}{x+1}+c$
7. $\displaystyle\int\sin(\ln x)\,\mathrm{d}x=\frac{x\left(\sin(\ln x)-\cos(\ln x)\right)}{2}+c$
8. $\displaystyle\int\frac{\ln x}{\sqrt{1+x}}\,\mathrm{d}x=\ln x\cdot 2\sqrt{1+x}-4\sqrt{1+x}-2\ln(\sqrt{1+x}-1)+2\ln(\sqrt{1+x}+1)+c$

解.

1. 令 $u=\ln x$, 則 $\mathrm{d}u=\dfrac{1}{x}\mathrm{d}x$; $\mathrm{d}v=x^3\,\mathrm{d}x$, 則 $v=\dfrac{x^4}{4}$. 故 $\displaystyle\int x^3\ln x\,\mathrm{d}x=\ln x\cdot\frac{x^4}{4}-\int\frac{x^4}{4}\cdot\frac{1}{x}\mathrm{d}x=\ln x\cdot\frac{x^4}{4}-\int\frac{x^3}{4}\,\mathrm{d}x=\ln x\cdot\frac{x^4}{4}-\frac{x^4}{16}+c.$
2. 令 $u=(\ln x)^3$, 則 $\mathrm{d}u=3(\ln x)^2\cdot\dfrac{1}{x}\mathrm{d}x$; $\mathrm{d}v=x\,\mathrm{d}x$, 則 $v=\dfrac{x^2}{2}$. 故 $\displaystyle\int x(\ln x)^3\,\mathrm{d}x=(\ln x)^3\cdot\frac{x^2}{2}-\int\frac{x^2}{2}\cdot 3(\ln x)^2\cdot\frac{1}{x}\,\mathrm{d}x=(\ln x)^3\cdot\frac{x^2}{2}-\frac{3}{2}\int x(\ln x)^2\,\mathrm{d}x$. 令 $u=(\ln x)^2$, 則 $\mathrm{d}u=2\ln x\cdot\dfrac{1}{x}\mathrm{d}x$; $\mathrm{d}v=x\,\mathrm{d}x$, 則 $v=\dfrac{x^2}{2}$.

故 $\int x(\ln x)^2\,\mathrm{d}x = (\ln x)^2\cdot\frac{x^2}{2} - \int \frac{x^2}{2}\cdot 2\ln x\cdot\frac{1}{x}\,\mathrm{d}x = (\ln x)^2\cdot\frac{x^2}{2} - \int x\ln x\,\mathrm{d}x$. 令 $u=\ln x$, 則 $\mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$; $\mathrm{d}v = x\,\mathrm{d}x$, 則 $v=\frac{x^2}{2}$. 故 $\int x\ln x\,\mathrm{d}x = \ln x\cdot\frac{x^2}{2} - \int\frac{x^2}{2}\cdot\frac{1}{x}\,\mathrm{d}x = \ln x\cdot\frac{x^2}{2} - \frac{1}{2}\int x\,\mathrm{d}x = \ln x\cdot\frac{x^2}{2} - \frac{x^2}{4}$. 以上, $\int x(\ln x)^3\,\mathrm{d}x = (\ln x)^3\cdot\frac{x^2}{2} - \frac{3}{2}\Big((\ln x)^2\cdot\frac{x^2}{2} - \big(\ln x\cdot\frac{x^2}{2} - \frac{x^2}{4}\big)\Big) = \frac{x^2}{2}\big((\ln x)^3 - \frac{3(\ln x)^2}{2} + \frac{3\ln x}{2} - \frac{3}{4}\big) + c$

3. 令 $u=\tan^{-1}x$, 則 $\mathrm{d}u = \frac{1}{1+x^2}\,\mathrm{d}x$; $\mathrm{d}v = x^2\,\mathrm{d}x$, 則 $v=\frac{x^3}{3}$. 故 $\int x^2\tan^{-1}x\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^3}{3} - \int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x$. 又 $\int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x = \frac{1}{3}\int\frac{(x^3+x)-x}{x^2+1}\,\mathrm{d}x = \frac{1}{3}\int\frac{x(x^2+1)-x}{x^2+1}\,\mathrm{d}x = \frac{1}{3}\int\big(x - \frac{x}{x^2+1}\big)\,\mathrm{d}x$. 令 $w=x^2+1$, 則 $\mathrm{d}w = 2x\,\mathrm{d}x \implies x\,\mathrm{d}x = \frac{1}{2}\,\mathrm{d}w$, 故 $\int\frac{x}{x^2+1}\,\mathrm{d}x = \int\frac{1}{x^2+1}\cdot x\,\mathrm{d}x = \int\frac{1}{w}\cdot\frac{1}{2}\,\mathrm{d}w = \frac{1}{2}\ln w + c = \frac{1}{2}\ln(x^2+1)+c$. 以上, $\int x^2\tan^{-1}x\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^3}{3} - \int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^3}{3} - \frac{x^2}{6} + \frac{\ln(x^2+1)}{6} + c$.

4. 令 $w=x^2$, 則 $\mathrm{d}w = 2x\,\mathrm{d}x \implies x\,\mathrm{d}x = \frac{1}{2}\,\mathrm{d}w$, 故 $\int x^5e^{-x^2}\,\mathrm{d}x = \int e^{-x^2}\cdot(x^2)^2\cdot x\,\mathrm{d}x = \int e^{-w}\cdot w^2\cdot\frac{1}{2}\,\mathrm{d}w = \frac{1}{2}\int w^2\,e^{-w}\,\mathrm{d}w = -\frac{1}{2}\,e^{-w}(w^2+2w+2) = -\frac{1}{2}\,e^{-x^2}(x^4+2x^2+2)+c$.

Diff		Int
w^2	$+$	e^{-w}
$2w$	$-$	$-e^{-w}$
2	$+$	e^{-w}
0		$-e^{-w}$

$$\int w^2e^{-w}\,\mathrm{d}w = -w^2\,e^{-w} - 2w\,e^{-w} - 2\,e^{-w} = -e^{-w}(w^2+2w+2)$$

5. 令 $w=\sqrt{x}$, 則 $w^2=x$, $\mathrm{d}x = 2w\,\mathrm{d}w$, 故 $\int xe^{\sqrt{x}}\,\mathrm{d}x = \int w^2e^w\cdot 2w\,\mathrm{d}w = 2\int w^3e^w\,\mathrm{d}w = 2\,e^w(w^3 - 3w^2+6w-6) = 2\,e^{\sqrt{x}}\,(x\sqrt{x} - 3x + 6\sqrt{x} - 6) + c$

Diff		Int
w^3	$+$	e^w
$3w^2$	$-$	e^w
$6w$	$+$	e^w
6	$-$	e^w
0		e^w

$$\int w^3\,e^w\,\mathrm{d}w = w^3\,e^w - 3w^2\,e^w + 6w\,e^w - 6\,e^w = e^w(w^3 - 3w^2 + 6w - 6)$$

6. $\int\frac{xe^x}{(x+1)^2}\,\mathrm{d}x = \int\frac{(x+1-1)e^x}{(x+1)^2}\,\mathrm{d}x = \int\frac{e^x}{x+1}\,\mathrm{d}x - \int\frac{e^x}{(x+1)^2}\,\mathrm{d}x$. 令 $u=e^x$, 則 $\mathrm{d}u = e^x\,\mathrm{d}x$; $\mathrm{d}v = \frac{1}{(x+1)^2}\,\mathrm{d}x$, 則 $v=\frac{-1}{x+1}$. 故 $\int\frac{e^x}{(x+1)^2}\,\mathrm{d}x = e^x\cdot\frac{-1}{x+1} + \int\frac{1}{x+1}\cdot e^x\,\mathrm{d}x = \frac{-e^x}{x+1} + \int\frac{e^x}{x+1}\,\mathrm{d}x$; 原式 $\int\frac{xe^x}{(x+1)^2}\,\mathrm{d}x = \int\frac{e^x}{x+1}\,\mathrm{d}x - \int\frac{e^x}{(x+1)^2}\,\mathrm{d}x = \int\frac{e^x}{x+1}\,\mathrm{d}x - \Big(\frac{-e^x}{x+1} + \int\frac{e^x}{x+1}\,\mathrm{d}x\Big) = \frac{e^x}{x+1}$.

7. 令 $w=\ln x$, 則 $e^w = x$, $\mathrm{d}x = e^w\,\mathrm{d}w$, 故 $\int\sin(\ln x)\,\mathrm{d}x = \int\sin w\cdot e^w\,\mathrm{d}w = \int e^w\,\sin w\,\mathrm{d}w = \frac{e^w(\sin w - \cos w)}{2} = \frac{x\,(\sin(\ln x) - \cos(\ln x))}{2} + c$

Diff		Int
$\sin w$	$+$	e^w
$\cos w$	$-$	e^w
$-\sin w$	$+$	e^w

$$\int e^w \sin w \,\mathrm{d}w = e^w \sin w - e^w \cos w - \int e^w \sin w \,\mathrm{d}w$$

$$\Longrightarrow \int e^w \sin w \,\mathrm{d}w = \frac{e^w(\sin w - \cos w)}{2}$$

8. 令 $u = \ln x$, 則 $\mathrm{d}u = \dfrac{1}{x}\,\mathrm{d}x$; $\mathrm{d}v = \dfrac{1}{\sqrt{1+x}}\,\mathrm{d}x$, 則 $v = 2\sqrt{1+x}$. 故 $\displaystyle\int \frac{\ln x}{\sqrt{1+x}}\,\mathrm{d}x = \ln x \cdot 2\sqrt{1+x} - 2\int \frac{\sqrt{1+x}}{x}\,\mathrm{d}x$. 令 $w = \sqrt{1+x}$, 則 $w^2 = 1+x \implies x = w^2 - 1$, $2w\,\mathrm{d}w = \mathrm{d}x$, 故 $\displaystyle\int \frac{\sqrt{1+x}}{x}\,\mathrm{d}x = \int \frac{w}{w^2-1} \cdot 2w\,\mathrm{d}w = 2\int \frac{w^2}{w^2-1}\,\mathrm{d}w = 2\int \frac{(w^2-1)+1}{w^2-1}\,\mathrm{d}w = 2w + 2\int \frac{1}{w^2-1}\,\mathrm{d}w = 2\sqrt{1+x} + 2\int \frac{1}{w^2-1}\,\mathrm{d}w$. 由 $\dfrac{1}{w^2-1} = \dfrac{1}{2}\left(\dfrac{1}{w-1} - \dfrac{1}{w+1}\right)$, $\displaystyle\int \frac{1}{w^2-1}\,\mathrm{d}w = \frac{1}{2}\int\left(\frac{1}{w-1} - \frac{1}{w+1}\right)\mathrm{d}w = \frac{1}{2}\left(\ln|w-1| - \ln|w+1|\right) = \frac{1}{2}\left(\ln|\sqrt{1+x}-1| - \ln|\sqrt{1+x}+1|\right)$. 以上, $\displaystyle\int \frac{\ln x}{\sqrt{1+x}}\,\mathrm{d}x = \ln x \cdot 2\sqrt{1+x} - 2\left(2\sqrt{1+x} + 2\left(\frac{1}{2}\left(\ln|\sqrt{1+x}-1| - \ln|\sqrt{1+x}+1|\right)\right)\right) = \ln x \cdot 2\sqrt{1+x} - 4\sqrt{1+x} - 2\ln|\sqrt{1+x}-1| + 2\ln|\sqrt{1+x}+1| + c$.

4.2 定積分

定積分 ≈ (帶符號) 面積: x 軸上方為正, 下方為負.

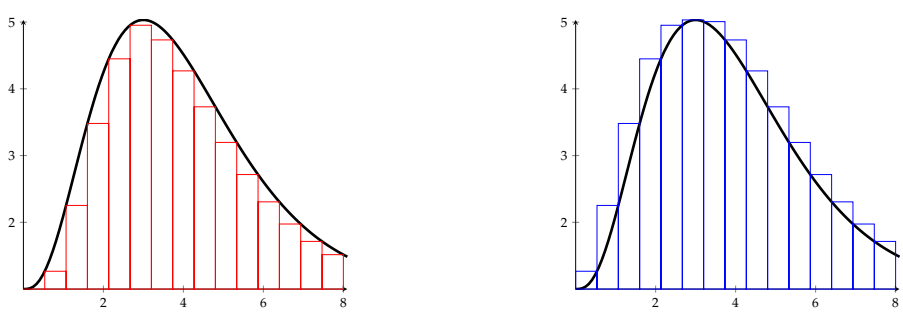
定義. 給定 $f : [a,\, b] \to \mathbb{R}$.

- $[a,\, b]$ 分割 $\mathbb{P} : a = x_0 < x_1 < x_2 < \cdots < x_n = b$
- $\Delta x_k = x_k - x_{k-1},\ k = 1,\, 2,\, \ldots,\, n;\ \|\mathbb{P}\| = \max\{\, |\Delta x_k| \mid 1 \leqslant k \leqslant n\}$
- 樣本點 ξ_k: $x_{k-1} \leqslant \xi_k \leqslant x_k,\ k = 1,\, 2,\, \ldots,\, n$
- $u_k = \sup\{\, f(x) \mid x_{k-1} \leqslant x \leqslant x_k\},\ l_k = \inf\{\, f(x) \mid x_{k-1} \leqslant x \leqslant x_k\},\ k = 1,\, 2,\, \ldots,\, n$
- $R(f, \mathbb{P}) = \displaystyle\sum_{k=1}^{n} f(\xi_k)\, \Delta x_k,\ U(f, \mathbb{P}) = \sum_{k=1}^{n} u_k\, \Delta x_k,\ L(f, \mathbb{P}) = \sum_{k=1}^{n} l_k\, \Delta x_k;$
  顯然 $L(f, \mathbb{P}) \leqslant R(f, \mathbb{P}) \leqslant U(f, \mathbb{P})$.
- 求 $\displaystyle\lim_{\|\mathbb{P}\| \to 0} R(f, \mathbb{P})$. 若對不同分割與樣本點選取此極限均存在且相等, 稱 f 在 $[a, b]$ 可積 (分) ; $f(x)$ 在 $[a,\, b]$ 的定積分 $\displaystyle\int_a^b f(x)\, \mathrm{d}x \equiv \lim_{\|\mathbb{P}\| \to 0} R(f, \mathbb{P})$

註.

- 在 $\displaystyle\int_a^b f(x)\, \mathrm{d}x$ 中, a 為*積分下限* (lower limit of integration) , b 為*積分上限* (upper limit of integration) , $f(x)$ 為*被積分式* (integrand) , x 為*積分變數* (variable of integration) .
- $\displaystyle\int_a^b f(x)\, \mathrm{d}x = \int_a^b f(t)\, \mathrm{d}t = \int_a^b f(u)\, \mathrm{d}u$ (定積分數值與積分變數無關)

結論. 若 f 在 $[a,\, b]$ 連續, 則 f 在 $[a,\, b]$ 可積.



性質. 令 $f,\, g$ 在包含 $a,\, b,\, c$ 之區間為可積, $\alpha,\, \beta \in \mathbb{R}$. 則

1. $\displaystyle\int_a^a f(x)\, \mathrm{d}x = 0$
2. $\displaystyle\int_a^b f(x)\, \mathrm{d}x = -\int_b^a f(x)\, \mathrm{d}x$
3. $\displaystyle\int_a^b \big(\alpha\, f(x) + \beta\, g(x)\big)\, \mathrm{d}x = \alpha \int_a^b f(x)\, \mathrm{d}x + \beta \int_a^b g(x)\, \mathrm{d}x$
4. $\displaystyle\int_a^b f(x)\, \mathrm{d}x + \int_b^c f(x)\, \mathrm{d}x = \int_a^c f(x)\, \mathrm{d}x$
5. $\displaystyle\int_a^b f(x)\, \mathrm{d}x \leqslant \int_a^b g(x)\, \mathrm{d}x$, 若 $f(x) \leqslant g(x)\ \forall\, x \in [a, b],\ a \leqslant b$

6. $\left|\int_a^b f(x)\,\mathrm{d}x\right| \leqslant \int_a^b |\,f(x)\,|\,\mathrm{d}x, \quad a \leqslant b$

7. $f(x)$ 為奇函數: $\int_{-a}^{a} f(x)\,\mathrm{d}x = 0$

8. $f(x)$ 為偶函數: $\int_{-a}^{a} f(x)\,\mathrm{d}x = 2\int_0^a f(x)\,\mathrm{d}x$

例.

1. $\int_{-\sqrt{2}}^{\sqrt{2}} \sqrt{2-x^2}\,\mathrm{d}x = \pi$ (半徑 $\sqrt{2}$ 之半圓面積)

2. $\int_{-1}^{2} \operatorname{sgn}(x)\,\mathrm{d}x = 2\cdot 1 - 1\cdot 1 = 1$; $\operatorname{sgn}(x) \equiv \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$

3. $\int_{-\pi}^{\pi} \sin(x^7 - 5x^3)\,\mathrm{d}x = 0$ ($\sin(x^7-5x^3)$ 為奇函數)

4. $\int_{-6}^{6} e^{-x^4}\sin(\sin x)\,\mathrm{d}x = 0$ ($e^{-x^4}\sin(\sin x)$ 為奇函數)

5. $\int_{-4}^{4} \left(e^x - e^{-x}\right)\mathrm{d}x = 0$ ($e^x - e^{-x}$ 為奇函數)

6. $\int_{-2024}^{2024} \left(e^{9x^5-2x^7} - e^{-9x^5+2x^7}\right)\mathrm{d}x = 0$ ($e^{9x^5-2x^7} - e^{-9x^5+2x^7}$ 為奇函數)

7. $\int_{-a}^{a} |x|\,\mathrm{d}x = 2\int_0^a |x|\,\mathrm{d}x = a^2$ ($|x|$ 為偶函數; 兩 $a\times a$ 等腰直角三角形面積)

8. 定義 $\int_1^x \frac{1}{\tau}\,\mathrm{d}\tau \equiv \ln x$, 則 $\int_{\frac{1}{4}}^{3} \frac{1}{x}\,\mathrm{d}x = \int_{\frac{1}{4}}^{1} \frac{1}{x}\,\mathrm{d}x + \int_1^3 \frac{1}{x}\,\mathrm{d}x = -\int_1^{\frac{1}{4}} \frac{1}{x}\,\mathrm{d}x + \int_1^3 \frac{1}{x}\,\mathrm{d}x = \ln 12$

9. $\int_0^2 \sqrt{4-x^2}\cdot\operatorname{sgn}(1-x)\,\mathrm{d}x = \int_0^1 \sqrt{4-x^2}\,\mathrm{d}x - \int_1^2 \sqrt{4-x^2}\,\mathrm{d}x = \left(\frac{4\pi}{12} + \frac{\sqrt{3}}{2}\right) - \left(\frac{4\pi}{6} - \frac{\sqrt{3}}{2}\right) = \sqrt{3} - \frac{\pi}{3}$

10. $\forall\, n \in \mathbb{N}$, $\int_n^{n+1} \lfloor x \rfloor\,\mathrm{d}x = n$

以極限定義求定積分

結論. $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

證. 由 $k(k+1) = \frac{k(k+1)(k+2) - (k-1)k(k+1)}{3}$, $\sum_{k=1}^{n} k(k+1) = \frac{n(n+1)(n+2)}{3}$. 又 $k^2 = k(k+1) - k$,

$$\sum_{k=1}^{n} k^2 = \sum_{k=1}^{n} \big(k(k+1) - k\big) = \frac{n(n+1)(n+2)}{3} - \frac{n(n+1)}{2} = \frac{n(n+1)(2n+1)}{6}.$$

例. 求 $\int_0^1 x^2\,\mathrm{d}x$.

解. 建立 $[0,\,1]$ 分割 $\mathbb{P}:\left\{0,\frac{1}{n},\frac{2}{n},\ldots,\frac{n}{n}\right\}$, 則 $\Delta x_k=\frac{1}{n}\ \forall\, k=1,2,\ldots,n$ 且 $\|P\|\to 0$ 當 $n\to\infty$. $f(x)=x^2$ 且 f 在 $[0,\,1]$ 嚴格遞增, $U(f,\mathbb{P})=\sum_{k=1}^{n}u_k\,\Delta x_k=\sum_{k=1}^{n}\frac{k^2}{n^2}\cdot\frac{1}{n}=\frac{1}{n^3}\sum_{k=1}^{n}k^2$, $L(f,\mathbb{P})=\sum_{k=1}^{n}l_k\,\Delta x_k=\sum_{k=1}^{n}\frac{(k-1)^2}{n^2}\cdot\frac{1}{n}=\frac{1}{n^3}\sum_{k=1}^{n-1}k^2$, $L(f,\mathbb{P})\leqslant R(f,\mathbb{P})\leqslant U(f,\mathbb{P})$. 由 $\sum_{k=1}^{n}k^2=\frac{n(n+1)(2n+1)}{6}$, $\lim_{n\to\infty}U(f,\mathbb{P})=\lim_{n\to\infty}\frac{1}{n^3}\sum_{k=1}^{n}k^2=\lim_{n\to\infty}\frac{n(n+1)(2n+1)}{6n^3}=\frac{1}{3}$, $\lim_{n\to\infty}L(f,\mathbb{P})=\lim_{n\to\infty}\frac{1}{n^3}\sum_{k=1}^{n-1}k^2=\lim_{n\to\infty}\frac{(n-1)n(2n-1)}{6n^3}=\frac{1}{3}$, 故由夾擊定理 $\lim_{n\to\infty}R(f,\mathbb{P})=\frac{1}{3}$.

結論. 若 $r\in\mathbb{R}$, $\sum_{k=0}^{n}r^k=\frac{r^{n+1}-1}{r-1}$.

證. 令 $s=\sum_{k=0}^{n}r^k$, 則 $r\,s=\sum_{k=1}^{n+1}r^k$; $r\,s-s=r^{n+1}-1\implies s=\frac{r^{n+1}-1}{r-1}$.

例. 求 $\int_0^1 e^x\,\mathrm{d}x$.

解. 建立 $[0,\,1]$ 分割 $\mathbb{P}:\left\{0,\frac{1}{n},\frac{2}{n},\ldots,\frac{n}{n}\right\}$, 則 $\Delta x_k=\frac{1}{n}\ \forall\, k=1,2,\ldots,n$ 且 $\|P\|\to 0$ 當 $n\to\infty$. $f(x)=e^x$ 且 f 在 $[0,\,1]$ 嚴格遞增, $U(f,\mathbb{P})=\sum_{k=1}^{n}u_k\,\Delta x_k=\sum_{k=1}^{n}e^{\frac{k}{n}}\cdot\frac{1}{n}=\frac{1}{n}\frac{e^{\frac{1}{n}}\left(e^{\frac{n}{n}}-1\right)}{e^{\frac{1}{n}}-1}=\frac{e-1}{n}\frac{e^{\frac{1}{n}}}{e^{\frac{1}{n}}-1}$, $L(f,\mathbb{P})=\sum_{k=1}^{n}l_k\,\Delta x_k=\sum_{k=1}^{n}e^{\frac{k-1}{n}}\cdot\frac{1}{n}=\frac{1}{n}\frac{e^{0}\left(e^{\frac{n}{n}}-1\right)}{e^{\frac{1}{n}}-1}=\frac{e-1}{n}\frac{1}{e^{\frac{1}{n}}-1}$, $L(f,\mathbb{P})\leqslant R(f,\mathbb{P})\leqslant U(f,\mathbb{P})$. $\lim_{n\to\infty}U(f,\mathbb{P})=\lim_{n\to\infty}\frac{e-1}{n}\frac{e^{\frac{1}{n}}}{e^{\frac{1}{n}}-1}=(e-1)\lim_{x\to 0+}\frac{xe^x}{e^x-1}=(e-1)\lim_{x\to 0+}\frac{e^x+xe^x}{e^x}=e-1$, $\lim_{n\to\infty}L(f,\mathbb{P})=\lim_{n\to\infty}\frac{e-1}{n}\frac{1}{e^{\frac{1}{n}}-1}=(e-1)\lim_{x\to 0+}\frac{x}{e^x-1}=(e-1)\lim_{x\to 0+}\frac{1}{e^x}=e-1$, 故由夾擊定理 $\lim_{n\to\infty}R(f,\mathbb{P})=e-1$.

結論. $\sum_{k=1}^{n}\cos kx=\frac{1}{2}\left(\frac{\sin\left(n+\frac{1}{2}\right)x}{\sin\frac{x}{2}}-1\right)$

證. 考慮 $\sum_{k=1}^{n}e^{ikx}=\frac{e^{ix}(1-e^{inx})}{1-e^{ix}}$, 則 $\sum_{k=1}^{n}\cos kx=\Re\left\{\sum_{k=1}^{n}e^{ikx}\right\}=\Re\left\{\frac{e^{ix}(1-e^{inx})}{1-e^{ix}}\right\}=\Re\left\{\frac{e^{ix}\,e^{\frac{inx}{2}}\left(e^{-\frac{inx}{2}}-e^{\frac{inx}{2}}\right)}{e^{\frac{ix}{2}}\left(e^{-\frac{ix}{2}}-e^{\frac{ix}{2}}\right)}\right\}=\frac{\cos\frac{(n+1)x}{2}\,\sin\frac{nx}{2}}{\sin\frac{x}{2}}=\frac{\sin\left(\frac{nx}{2}+\frac{(n+1)x}{2}\right)+\sin\left(\frac{nx}{2}+\frac{(n+1)x}{2}\right)}{2\,\sin\frac{x}{2}}=\frac{\sin\left(n+\frac{1}{2}\right)x-\sin\frac{x}{2}}{2\,\sin\frac{x}{2}}=\frac{1}{2}\left(\frac{\sin\left(n+\frac{1}{2}\right)x}{\sin\frac{x}{2}}-1\right)$.

例. 求 $\int_0^{\frac{\pi}{2}}\cos x\,\mathrm{d}x$.

解. 建立 $\left[0,\,\frac{\pi}{2}\right]$ 分割 $\mathbb{P}:\left\{0,\frac{\pi}{2n},\frac{2\pi}{2n},\ldots,\frac{n\pi}{2n}\right\}$, 則 $\Delta x_k=\frac{\pi}{2n}\ \forall\, k=1,2,\ldots,n$ 且 $\|P\|\to 0$ 當 $n\to\infty$, 則積分為 $\lim_{n\to\infty}R(f,\mathbb{P})=\sum_{k=1}^{n}f(\xi_k)\,\Delta x_k=\lim_{n\to\infty}\sum_{k=1}^{n}\cos\frac{k\pi}{2n}\cdot\frac{\pi}{2n}=\lim_{n\to\infty}\sum_{k=1}^{n}\cos\left(k\left(\frac{\pi}{2n}\right)\right)\cdot\frac{\pi}{2n}=\lim_{n\to\infty}\frac{1}{2}\left(\frac{\sin\left(\left(n+\frac{1}{2}\right)\cdot\frac{\pi}{2n}\right)}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}-1\right)\cdot\frac{\pi}{2n}=\lim_{n\to\infty}\frac{\sin\left(\left(n+\frac{1}{2}\right)\frac{\pi}{2n}\right)\frac{1}{2}\cdot\frac{\pi}{2n}}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}=\lim_{n\to\infty}\sin\left(\frac{1}{2}+\frac{1}{4n}\right)\pi\cdot\lim_{n\to\infty}\frac{\frac{1}{2}\cdot\frac{\pi}{2n}}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}=\sin\frac{\pi}{2}\cdot 1=1$.

微積分基本定理

定理 (微積分基本定理 (Fundamental Theorem of Calculus, FTC)).

1. 若 f 在 $[a,\, b]$ 連續, 令 $F(x) = \displaystyle\int_a^x f(\tau)\,\mathrm{d}\tau$ 且 $a \leqslant x \leqslant b$, 則 $F'(x) = f(x)\ \forall\, x \in [a,\, b]$.
2. 若 $G'(x) = f(x)\ \forall\, x \in [a,\, b]$, 則 $\displaystyle\int_a^b f(x)\,\mathrm{d}x = G(b) - G(a) \equiv G(x)\Big|_a^b$.

註. 由 FTC, $\displaystyle\int_a^b f(x)\,\mathrm{d}x$ 可由 f 的反導函數 (不定積分) 得出, 不需繁複極限計算!

證.

1. WLOG 令 $h > 0$. 則 $\left|\dfrac{F(x+h)-F(x)}{h} - f(x)\right| = \left|\dfrac{1}{h}\displaystyle\int_x^{x+h}(f(\tau)-f(x))\,\mathrm{d}\tau\right| \leqslant \dfrac{1}{h}\displaystyle\int_x^{x+h}\big|f(\tau) - f(x)\big|\,\mathrm{d}\tau \leqslant \max_{x\leqslant\tau\leqslant x+h}\big|f(\tau)-f(x)\big| \to 0$, 當 $h \to 0$ 且 f 為連續.
2. $(G(x)-F(x))' = f(x) - f(x) = 0\ \forall\, x \in [a,b]$, 故 $G(x)-F(x)$ 在 $[a,b]$ 為常數 $\implies G(a)-F(a) = G(b)-F(b) \implies G(b)-G(a) = F(b)-F(a) = \displaystyle\int_a^b f(x)\,\mathrm{d}x$.

例 (以 FTC 求定積分).

1. x^2 之反導函數為 $\dfrac{x^3}{3}$, 故 $\displaystyle\int_0^1 x^2\,\mathrm{d}x = \frac{1^3}{3} - \frac{0^3}{3} = \frac{1}{3}$.
2. e^x 之反導函數為 e^x, 故 $\displaystyle\int_0^1 e^x\,\mathrm{d}x = e^1 - e^0 = e - 1$.
3. $\cos x$ 之反導函數為 $\sin x$, 故 $\displaystyle\int_0^{\frac{\pi}{2}} \cos x\,\mathrm{d}x = \sin\frac{\pi}{2} - \sin 0 = 1$.

結論 (定積分變數變換).

- 求反導函數後代入: $\displaystyle\int_a^b f'(g(x))\,g'(x)\,\mathrm{d}x = f(g(x))\Big|_{x=a}^{x=b} = f(g(b)) - f(g(a))$
- 變數變換並改變積分範圍: $\displaystyle\int_a^b f'(g(x))\,g'(x)\,\mathrm{d}x = \int_a^b f'(g(x))\,\mathrm{d}g(x) = \int_{g(a)}^{g(b)} f'(u)\,\mathrm{d}u = f(u)\Big|_{u=g(a)}^{u=g(b)} = f(g(b)) - f(g(a))$

例. 求 $\displaystyle\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x$.

解.

- 求反導函數後代入: 令 $u = 1 + x^4$, 則 $\mathrm{d}u = 4x^3\,\mathrm{d}x \implies x^3\,\mathrm{d}x = \dfrac{\mathrm{d}u}{4}$, 故 $\displaystyle\int x^3(1+x^4)^3\,\mathrm{d}x = \int(1+x^4)^3\,x^3\,\mathrm{d}x = \int u^3\,\frac{\mathrm{d}u}{4} = \frac{u^4}{16} + c = \frac{(1+x^4)^4}{16} + c$. 故 $\displaystyle\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x = \frac{(1+x^4)^4}{16}\Big|_{x=0}^{x=1} = \frac{(1+1^4)^4 - (1+0^4)^4}{16} = \frac{15}{16}$.
- 變數變換並改變積分範圍: 令 $u = 1 + x^4$, 則 $\mathrm{d}u = 4x^3\,\mathrm{d}x \implies x^3\,\mathrm{d}x = \dfrac{\mathrm{d}u}{4}$. 積分範圍 x 由 0 至 1, 則變數變換後 u 由 $1 + 0^4 = 1$ 至 $1 + 1^4 = 2$, 故 $\displaystyle\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x = \int_0^1 (1+x^4)^3\,x^3\,\mathrm{d}x = \int_1^2 u^3\,\frac{\mathrm{d}u}{4} = \frac{1}{4}\int_1^2 u^3\,\mathrm{d}u = \frac{1}{16}u^4\Big|_{u=1}^{u=2} = \frac{2^4 - 1^4}{16} = \frac{15}{16}$.

例. 求 $\displaystyle\int_0^{\sqrt{3}} \frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x$.

解.

- 求反導函數後代入: 令 $u = 1+x^2$, 則 $\mathrm{d}u = 2x\,\mathrm{d}x \implies 4x\,\mathrm{d}x = 2\,\mathrm{d}u$, 故 $\displaystyle\int \frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x = \int \frac{2}{\sqrt{u}}\,\mathrm{d}u = 4\sqrt{u} + c = 4\sqrt{1+x^2} + c$. 故 $\displaystyle\int_0^{\sqrt{3}} \frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x = 4\sqrt{1+x^2}\Big|_{x=0}^{x=\sqrt{3}} = 4\sqrt{1+3} - 4\sqrt{1} = 4$.
- 變數變換並改變積分範圍: 令 $u = 1+x^2$, 則 $\mathrm{d}u = 2x\,\mathrm{d}x \implies 4x\,\mathrm{d}x = 2\,\mathrm{d}u$. 積分範圍 x 由 0 至 $\sqrt{3}$, 則變數變換後 u 由 $1+0^2 = 1$ 至 $1+(\sqrt{3})^2 = 4$, 故 $\displaystyle\int_0^{\sqrt{3}} \frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x = \int_1^4 \frac{2}{\sqrt{u}}\,\mathrm{d}u = 4\sqrt{u}\Big|_{u=1}^{u=4} = 4\left(\sqrt{4} - \sqrt{1}\right) = 4$.

例. 求 $\displaystyle\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x$.

解.

- 求反導函數後代入: 令 $u = \cos x$, 則 $\mathrm{d}u = -\sin x\,\mathrm{d}x \implies \sin x\,\mathrm{d}x = -\mathrm{d}u$, 故 $\displaystyle\int 3\cos^2 x \sin x\,\mathrm{d}x = -3\int u^2\,\mathrm{d}u = -u^3 + c = -\cos^3 x + c$. 故 $\displaystyle\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x = -\cos^3 x\Big|_{x=0}^{x=\pi} = -(\cos^3\pi - \cos^3 0) = -((-1)^3 - 1^3) = 2$.
- 變數變換並改變積分範圍: 令 $u = \cos x$, 則 $\mathrm{d}u = -\sin x\,\mathrm{d}x \implies \sin x\,\mathrm{d}x = -\mathrm{d}u$. 積分範圍 x 由 0 至 π, 則變數變換後 u 由 $\cos 0 = 1$ 至 $\cos\pi = -1$, 故 $\displaystyle\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x = -\int_1^{-1} 3u^2\,\mathrm{d}u = -u^3\Big|_{u=1}^{u=-1} = -((-1)^3 - 1^3) = 2$.

例. 若 f 在 $[a,b]$ 二次可微且 $f(a) = f(b) = 0$, 證明 $\displaystyle\int_a^b (x-a)(b-x)\,f''(x)\,\mathrm{d}x = -2\int_a^b f(x)\,\mathrm{d}x$.

解. $\displaystyle\int_a^b (x-a)(b-x)\,f''(x)\,\mathrm{d}x = \big((x-a)(b-x)\,f'(x) - (a+b-2x)\,f(x)\big)\Big|_a^b - 2\int_a^b f(x)\,\mathrm{d}x = \big((b-a)(b-b)\,f'(b) - (a+b-2b)\,f(b)\big) - \big((a-a)(b-a)\,f'(a) - (a+b-2a)\,f(a)\big) - 2\int_a^b f(x)\,\mathrm{d}x = -2\int_a^b f(x)\,\mathrm{d}x$.

Diff		Int
$(x-a)(b-x)$	$+$	$f''(x)$
$a+b-2x$	$-$	$f'(x)$
-2	$+$	$f(x)$
0		$\int f(x)\,\mathrm{d}x$

$$\int (x-a)(b-x)\,f''(x)\,\mathrm{d}x = \big((x-a)(b-x)\,f'(x) - (a+b-2x)\,f(x)\big) - 2\int f(x)\,\mathrm{d}x$$

性質. 令 $\displaystyle F(x) = \int_{v(x)}^{u(x)} f(\tau)\,\mathrm{d}\tau$, 則 $F'(x) = f(u(x))\cdot u'(x) - f(v(x))\cdot v'(x)$.

證. 令 $a \in \mathbb{R}$, $\displaystyle F(x) = \int_{v(x)}^{u(x)} f(\tau)\,\mathrm{d}\tau = \int_a^{u(x)} f(\tau)\,\mathrm{d}\tau - \int_a^{v(x)} f(\tau)\,\mathrm{d}\tau$. 令 $\displaystyle G(x) \equiv \int_a^x f(\tau)\,\mathrm{d}\tau$, 則 $G'(x) = f(x)$, $\displaystyle F(x) = \int_a^{u(x)} f(\tau)\,\mathrm{d}\tau - \int_a^{v(x)} f(\tau)\,\mathrm{d}\tau = G(u(x)) - G(v(x))$; 故 $F'(x) = (G(u(x)) - G(v(x)))' = G'(u(x))\cdot u'(x) - G'(v(x))\cdot v'(x) = f(u(x))\cdot u'(x) - f(v(x))\cdot v'(x)$.

例.

1. $F(x)=\int_1^x \frac{1}{1+\tau^4}\,\mathrm{d}\tau \implies F'(x)=\frac{1}{1+x^4}$
2. $F(x)=\int_2^{\sqrt{x}} \sin\tau\,\mathrm{d}\tau \implies F'(x)=\sin\sqrt{x}\cdot\frac{1}{2\sqrt{x}}$
3. $F(x)=\int_x^{2x} \tau^3\,\mathrm{d}\tau \implies F'(x)=(2x)^3\cdot 2-x^3\cdot 1=15x^3$
4. $F(x)=\int_{\sin x}^{\tan^{-1}x} e^{\tau^2}\,\mathrm{d}\tau \implies F'(x)=e^{(\tan^{-1}x)^2}\cdot\frac{1}{1+x^2}-e^{\sin^2 x}\cdot\cos x$

例. 若 $g(x)=\int_0^{\cos x}(1+\sin(t^2)),\mathrm{d}t$, $f(x)=\int_0^{g(x)}\frac{1}{\sqrt{1+t^3}}\,\mathrm{d}t$, 求 $f'\big(\frac{\pi}{2}\big)$.

解. $f'(x)=\frac{1}{\sqrt{1+g(x)^3}}\cdot g'(x)$, $g'(x)=(1+\sin(\cos^2 x))\cdot(-\sin x)$. 代入 $x=\frac{\pi}{2}$, $g\big(\frac{\pi}{2}\big)=0$, $g'\big(\frac{\pi}{2}\big)=-1$, 故 $f'\big(\frac{\pi}{2}\big)=-1$.

例. 若 $\int_0^{x^2} f(t)\,\mathrm{d}t=x\sin\pi x$, 求 $f'(9)$.

解. $\int_0^{x^2} f(t)\,\mathrm{d}t=x\sin\pi x$ 兩邊對 x 微分得 $f(x^2)\cdot 2x=\sin\pi x+x\cdot\pi\cos\pi x$. 兩邊再對 x 微分得 $(f'(x^2)\cdot 2x)\cdot 2x+f(x^2)\cdot 2=\pi\cos\pi x+\pi\cos\pi x-x\cdot\pi^2\sin\pi x$. 代入 $x=3$, 則 $(f'(9)\cdot 2\cdot 3)\cdot 2\cdot 3+f(9)\cdot 2=\pi\cos 3\pi+\pi\cos 3\pi-3\cdot\pi^2\sin 3\pi \implies f'(9)\cdot 36+f(9)\cdot 2=-2\pi$. 又 $f(3^2)\cdot(2\cdot 3)=\sin 3\pi+3\cdot\pi\cos 3\pi \implies f(9)=-\frac{\pi}{2}$, 故 $f'(9)=-\frac{\pi}{36}$.

例. 求函數 f 與 $a\in\mathbb{R}$ 使 $6+\int_a^x \frac{f(t)}{t^2}\,\mathrm{d}t=2\sqrt{x}$, $\forall\, x>0$.

解. $6+\int_a^x \frac{f(t)}{t^2}\,\mathrm{d}t=2\sqrt{x}$ 兩邊對 x 微分得 $\frac{f(x)}{x^2}=\frac{1}{\sqrt{x}} \implies f(x)=x^{\frac{3}{2}}$. 代入原式得 $6+\int_a^x \frac{t^{\frac{3}{2}}}{t^2}\,\mathrm{d}t=6+\int_a^x \frac{1}{\sqrt{t}}\,\mathrm{d}t=6+2\sqrt{t}\Big|_a^x=6+2\sqrt{x}-2\sqrt{a}=2\sqrt{x} \implies a=9$.

例. 求下列極限.

1. $\lim_{x\to 0}\frac{\int_0^x(\sec t-1)\,\mathrm{d}t}{x^3}$
2. $\lim_{x\to\infty}\frac{\int_{x^2}^0 e^{t-x^2}(2t^2+1)\,\mathrm{d}t}{x^4}$
3. $\lim_{x\to 0}\frac{1}{x}\int_0^{\tan x} f(u)(\sin x-\cos u)\,\mathrm{d}u$, f 為連續函數

解.

1. $\lim_{x\to 0}\frac{\int_0^x(\sec t-1)\,\mathrm{d}t}{x^3}=\lim_{x\to 0}\frac{\sec x-1}{3x^2}=\lim_{x\to 0}\frac{\sec x\tan x}{6x}=\lim_{x\to 0}\frac{\sec x\tan x\cdot\tan x+\sec x\cdot\sec^2 x}{6}=\frac{1}{6}$.
2. $\lim_{x\to\infty}\frac{\int_{x^2}^0 e^{t-x^2}(2t^2+1)\,\mathrm{d}t}{x^4}=\lim_{x\to\infty}\frac{\int_{x^2}^0 e^t(2t^2+1)\,\mathrm{d}t}{e^{x^2}x^4}=\lim_{x\to\infty}\frac{-e^{x^2}(2(x^2)^2+1)\cdot 2x}{e^{x^2}\,2x\cdot x^4+e^{x^2}\cdot 4x^3}=\lim_{x\to\infty}\frac{-(4x^5+2x)}{2x^5+4x^3}=-2$.

3. $\lim_{x\to 0}\frac{1}{x}\int_0^{\tan x} f(u)(\sin x-\cos u)\,\mathrm{d}u = \lim_{x\to 0}\frac{\int_0^{\tan x} f(u)(\sin x-\cos u)\,\mathrm{d}u}{x}$
$= \lim_{x\to 0}\frac{\sin x\int_0^{\tan x} f(u)\,\mathrm{d}u - \int_0^{\tan x} f(u)\cos u\,\mathrm{d}u}{x}$
$= \lim_{x\to 0}\frac{\cos x\int_0^{\tan x} f(u)\,\mathrm{d}u + \sin x\cdot f(\tan x)\cdot\sec^2 x - f(\tan x)\cos(\tan x)\cdot\sec^2 x}{1} = -f(0)$

4.3 積分技巧

部份分式

例 (動機)**.** 若 $a\neq 0$, 求 $\int\frac{1}{x^2-a^2}\,\mathrm{d}x$.

解. 由 $\frac{1}{x^2-a^2}=\frac{1}{2a}\left(\frac{1}{x-a}-\frac{1}{x+a}\right)$, $\int\frac{1}{x^2-a^2}\,\mathrm{d}x=\int\frac{1}{2a}\left(\frac{1}{x-a}-\frac{1}{x+a}\right)\mathrm{d}x=\frac{1}{2a}\big(\ln|x-a|-\ln|x+a|\big)$

結論.

1. 實係數多項式可分解成不可約的一次及二次因式的乘積.
2. 有理式可寫成多項式與真分式之和.
3. 若 $\frac{p(x)}{q(x)}$ 為一真分式, $q(x)=(x+a_1)^{m_1}(x+a_2)^{m_2}\cdots(x+a_k)^{m_k}\cdot(x^2+b_1x+c_1)^{n_1}(x^2+b_2x+c_2)^{n_2}\cdots(x^2+b_lx+c_l)^{n_l}$, 其中 $(x+a_i)$, $(x^2+b_ix+c_i)$ 均相異, $(x^2+b_ix+c_i)$ 為不可分解之二次式 $(b_i^2-4c_i<0)$, 則

$$\begin{aligned}\frac{p(x)}{q(x)} &= \frac{\alpha_{11}}{x+a_1}+\frac{\alpha_{12}}{(x+a_1)^2}+\cdots+\frac{\alpha_{1m_1}}{(x+a_1)^{m_1}}+\\ &\quad \frac{\alpha_{21}}{x+a_2}+\frac{\alpha_{22}}{(x+a_2)^2}+\cdots+\frac{\alpha_{2m_2}}{(x+a_2)^{m_2}}+\cdots+\\ &\quad \frac{\alpha_{k1}}{x+a_k}+\frac{\alpha_{k2}}{(x+a_k)^2}+\cdots+\frac{\alpha_{km_k}}{(x+a_k)^{m_k}}+\\ &\quad \frac{\beta_{11}x+\gamma_{11}}{x^2+b_1x+c_1}+\frac{\beta_{12}x+\gamma_{12}}{(x^2+b_1x+c_1)^2}+\cdots+\frac{\beta_{1n_1}x+\gamma_{1n_1}}{(x^2+b_1x+c_1)^{n_1}}+\\ &\quad \frac{\beta_{21}x+\gamma_{21}}{x^2+b_2x+c_2}+\frac{\beta_{22}x+\gamma_{22}}{(x^2+b_2x+c_2)^2}+\cdots+\frac{\beta_{2n_2}x+\gamma_{2n_2}}{(x^2+b_1x+c_1)^{n_2}}+\cdots+\\ &\quad \frac{\beta_{l1}x+\gamma_{l1}}{x^2+b_lx+c_l}+\frac{\beta_{l2}x+\gamma_{l2}}{(x^2+b_lx+c_l)^2}+\cdots+\frac{\beta_{ln_l}x+\gamma_{ln_l}}{(x^2+b_lx+c_l)^{n_l}}\end{aligned}$$

其中 α_{ij}, β_{ij}, $\gamma_{ij}\in\mathbb{R}$.

註. 任一有理函數之積分可分解為多項式積分與以下型式積分:

1. $\int\frac{1}{(x+a)^n}\,\mathrm{d}x$, $n\in\mathbb{N}$
2. $\int\frac{1}{(x^2+bx+c)^n}\,\mathrm{d}x$, $b^2-4c<0$, $n\in\mathbb{N}$
3. $\int\frac{x}{(x^2+bx+c)^n}\,\mathrm{d}x$, $b^2-4c<0$, $n\in\mathbb{N}$

例. 求 $\int\frac{x}{x^2-5x+6}\,\mathrm{d}x$.

解. $\dfrac{x}{x^2-5x+6}=\dfrac{x}{(x-3)(x-2)}=\dfrac{A}{x-3}+\dfrac{B}{x-2} \implies x=A\,(x-2)+B\,(x-3)$. 代入 $x=3 \implies 3=A$; 代入 $x=2 \implies 2=B(2-3) \implies B=-2$, 故 $\dfrac{x}{x^2-5x+6}=\dfrac{3}{x-3}-\dfrac{2}{x-2}$, $\displaystyle\int\frac{x}{x^2-5x+6}\,\mathrm{d}x=\int\left(\frac{3}{x-3}-\frac{2}{x-2}\right)\mathrm{d}x=3\ln|x-3|-2\ln|x-2|$

例. 求 $\displaystyle\int\frac{1}{x^3+1}\,\mathrm{d}x$.

解. $\dfrac{1}{x^3+1}=\dfrac{1}{(x+1)(x^2-x+1)}=\dfrac{A}{x+1}+\dfrac{Bx+C}{x^2-x+1} \implies 1=A\,(x^2-x+1)+(Bx+C)\,(x+1)$. 代入 $x=-1 \implies 1=A((-1)^2-(-1)+1)=3A \implies A=\dfrac{1}{3}$; 代入 $x=0 \implies 1=A+C \implies C=\dfrac{2}{3}$; 代入 $x=1 \implies 1=A+(B+C)\cdot 2 \implies 1=\dfrac{1}{3}+\left(B+\dfrac{2}{3}\right)\cdot 2 \implies B=-\dfrac{1}{3}$; 故 $\dfrac{1}{x^3+1}=\dfrac{1}{3}\left(\dfrac{1}{x+1}-\dfrac{x-2}{x^2-x+1}\right)$, $\displaystyle\int\frac{1}{x^3+1}\,\mathrm{d}x=\int\frac{1}{3}\left(\frac{1}{x+1}-\frac{x-2}{x^2-x+1}\right)\mathrm{d}x=\int\frac{1}{3}\left(\frac{1}{x+1}-\frac{\frac{1}{2}\,(2x-1)-\frac{3}{2}}{x^2-x+1}\right)\mathrm{d}x=\frac{1}{3}\int\frac{1}{x+1}\,\mathrm{d}x-\frac{1}{6}\int\frac{2x-1}{x^2-x+1}\,\mathrm{d}x-\frac{1}{2}\int\frac{1}{x^2-x+1}\,\mathrm{d}x=\frac{1}{3}\ln|x+1|-\frac{1}{6}\ln(x^2-x+1)-\frac{1}{\sqrt{3}}\tan^{-1}\frac{2x-1}{\sqrt{3}}$

例. 求 $\displaystyle\int\frac{1}{\cos^3 x}\,\mathrm{d}x$.

解. $\displaystyle\int\frac{1}{\cos^3 x}\,\mathrm{d}x=\int\frac{\cos x}{\cos^4 x}\,\mathrm{d}x=\int\frac{\cos x}{(1-\sin^2 x)^2}\,\mathrm{d}x$. 令 $u=\sin x$, 則 $\mathrm{d}u=\cos x\,\mathrm{d}x$, 則 $\displaystyle\int\frac{\cos x}{(1-\sin^2 x)^2}\,\mathrm{d}x=\int\frac{1}{(1-u^2)^2}\,\mathrm{d}u$. $\dfrac{1}{(1-u^2)^2}=\dfrac{1}{(u-1)^2(u+1)^2}=\dfrac{A}{u-1}+\dfrac{B}{(u-1)^2}+\dfrac{C}{u+1}+\dfrac{D}{(u+1)^2} \implies 1=A\,(u-1)(u+1)^2+B\,(u+1)^2+C\,(u-1)^2(u+1)+D\,(u-1)^2$. 代入 $u=1 \implies 1=4B \implies B=\dfrac{1}{4}$; 代入 $u=-1 \implies 1=4D \implies D=\dfrac{1}{4}$; 代入 $u=0 \implies 1=-A+B+C+D \implies 1=-A+\dfrac{1}{4}+C+\dfrac{1}{4} \implies \dfrac{1}{2}=-A+C$; 代入 $u=2 \implies 1=9A+9B+3C+D \implies 1=9A+\dfrac{9}{4}+3C+\dfrac{1}{4} \implies A=-\dfrac{1}{4},\ C=\dfrac{1}{4}$; 則 $\dfrac{1}{(1-u^2)^2}=\dfrac{1}{4}\left(\dfrac{-1}{u-1}+\dfrac{1}{(u-1)^2}+\dfrac{1}{u+1}+\dfrac{1}{(u+1)^2}\right)$, $\displaystyle\int\frac{1}{(1-u^2)^2}\,\mathrm{d}u=\int\frac{1}{4}\left(\frac{-1}{u-1}+\frac{1}{(u-1)^2}+\frac{1}{u+1}+\frac{1}{(u+1)^2}\right)\mathrm{d}x=\frac{1}{4}\left(-\ln|u-1|-\frac{1}{u-1}+\ln|u+1|-\frac{1}{u+1}\right)=\frac{1}{4}\left(-\ln|\sin x-1|-\frac{1}{\sin x-1}+\ln|\sin x+1|-\frac{1}{\sin x+1}\right)$.

部份分式另解: $\dfrac{1}{(1-u^2)^2}=\left(\dfrac{1}{u^2-1}\right)^2=\left(\dfrac{1}{(u-1)(u+1)}\right)^2=\dfrac{1}{4}\left(\dfrac{1}{u-1}-\dfrac{1}{u+1}\right)^2=\dfrac{1}{4}\left(\dfrac{1}{(u-1)^2}-\dfrac{2}{(u-1)(u+1)}+\dfrac{1}{(u+1)^2}\right)=\dfrac{1}{4}\left(\dfrac{1}{(u-1)^2}-\left(\dfrac{1}{u-1}-\dfrac{1}{u+1}\right)+\dfrac{1}{(u+1)^2}\right)=\dfrac{1}{4}\left(\dfrac{-1}{u-1}+\dfrac{1}{(u-1)^2}+\dfrac{1}{u+1}+\dfrac{1}{(u+1)^2}\right)$.

三角函數代換

結論.

1. 遇 $\sqrt{a^2-x^2}$, 考慮 $x=a\sin\theta \implies \theta=\sin^{-1}\dfrac{x}{a}$, $\mathrm{d}x=a\cos\theta\,\mathrm{d}\theta$
2. 遇 $\sqrt{a^2+x^2}$, 考慮 $x=a\tan\theta \implies \theta=\tan^{-1}\dfrac{x}{a}$, $\mathrm{d}x=a\sec^2\theta\,\mathrm{d}\theta$

3. 遇 $\sqrt{x^2-a^2}$, 考慮 $x=a\sec\theta \implies \theta=\sec^{-1}\dfrac{x}{a}$, $\mathrm{d}x=a\sec\theta\,\tan\theta\,\mathrm{d}\theta$

4. 遇 $\sin x$, $\cos x$ 之有理式, 考慮 $u=\tan\dfrac{x}{2}$, 由以下化為 u 之有理式:

   - $\sin x=2\sin\dfrac{x}{2}\cos\dfrac{x}{2}=2\dfrac{u}{\sqrt{1+u^2}}\dfrac{1}{\sqrt{1+u^2}}=\dfrac{2u}{1+u^2}$
   - $\cos x=2\cos^2\dfrac{x}{2}-1=2\cdot\dfrac{1}{1+u^2}-1=\dfrac{1-u^2}{1+u^2}$
   - $\mathrm{d}u=\dfrac{1}{2}\sec^2\dfrac{x}{2}\,\mathrm{d}x \implies \mathrm{d}x=\dfrac{2}{1+u^2}\,\mathrm{d}u$

例. 若 $a\neq 0$, 求下列不定積分.

1. $\displaystyle\int\sqrt{a^2-x^2}\,\mathrm{d}x$
2. $\displaystyle\int\frac{1}{x^2-a^2}\,\mathrm{d}x$
3. $\displaystyle\int\frac{1}{x^2\sqrt{x^2+a^2}}\,\mathrm{d}x$
4. $\displaystyle\int\sqrt{x^2+a^2}\,\mathrm{d}x$
5. $\displaystyle\int\frac{1}{\sqrt{x^2-a^2}}\,\mathrm{d}x$
6. $\displaystyle\int\frac{1}{a+\sin x}\,\mathrm{d}x$, $a>1$

解.

1. 令 $x=a\sin\theta$, 則 $\displaystyle\int\sqrt{a^2-x^2}\,\mathrm{d}x=\int\sqrt{a^2-a^2\sin^2\theta}\,a\cos\theta\,\mathrm{d}\theta=a^2\int\cos^2\theta\,\mathrm{d}\theta=a^2\int\frac{1+\cos 2\theta}{2}\,\mathrm{d}\theta$
$\displaystyle=\frac{a^2}{2}\theta+\frac{a^2}{4}\sin 2\theta=\frac{a^2}{2}\sin^{-1}\frac{x}{a}+\frac{a^2}{2}\frac{x}{a}\cdot\frac{\sqrt{a^2-x^2}}{a}=\frac{a^2}{2}\sin^{-1}\frac{x}{a}+\frac{x}{2}\sqrt{a^2-x^2}.$

2. 令 $x=a\sec\theta$, 則 $\displaystyle\int\frac{1}{x^2-a^2}\,\mathrm{d}x=\int\frac{a\sec\theta\tan\theta}{a^2\tan^2\theta}\,\mathrm{d}\theta=\frac{1}{a}\int\frac{1}{\sin\theta}\,\mathrm{d}\theta=\frac{1}{a}\int\csc\theta\,\mathrm{d}\theta$
$\displaystyle=\frac{1}{a}\int\csc\theta\,\frac{\csc\theta+\cot\theta}{\csc\theta+\cot\theta}\,\mathrm{d}\theta=-\frac{1}{a}\int\frac{\mathrm{d}(\csc\theta+\cot\theta)}{\csc\theta+\cot\theta}=-\frac{1}{a}\ln|\csc\theta+\cot\theta|=-\frac{1}{a}\ln\left|\frac{x}{\sqrt{x^2-a^2}}+\frac{a}{\sqrt{x^2-a^2}}\right|=-\frac{1}{a}\ln\left|\frac{\sqrt{x+a}}{\sqrt{x-a}}\right|=\frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|.$

3. 令 $x=a\tan\theta$, 則 $\displaystyle\int\frac{1}{x^2\sqrt{x^2+a^2}}\,\mathrm{d}x=\int\frac{a\sec^2\theta}{a^2\tan^2 x\cdot a\sec\theta}\,\mathrm{d}\theta=\int\frac{\sec\theta}{a^2\tan^2\theta}\,\mathrm{d}\theta=\int\frac{\cos\theta}{a^2\sin^2\theta}\,\mathrm{d}\theta$
$\displaystyle=-\frac{1}{a^2\sin\theta}=-\frac{\sqrt{x^2+a^2}}{a^2x}.$

4. 令 $x=a\tan\theta$, 則 $\displaystyle\int\sqrt{x^2+a^2}\,\mathrm{d}x=\int a\sec\theta\cdot a\sec^2\theta\,\mathrm{d}\theta=a^2\int\sec^3\theta\,\mathrm{d}\theta$. 令 $u=\sec\theta$, 則 $\mathrm{d}u=\sec\theta\tan\theta\,\mathrm{d}\theta$; 令 $\mathrm{d}v=\sec^2\theta\,\mathrm{d}\theta$, 則 $v=\tan\theta$. 故 $\displaystyle\int\sec^3\theta\,\mathrm{d}\theta=\sec\theta\cdot\tan\theta-\int\tan\theta\cdot\sec\theta\tan\theta\,\mathrm{d}\theta=\sec\theta\cdot\tan\theta-\int\sec\theta\cdot\tan^2\theta\,\mathrm{d}\theta=\sec\theta\cdot\tan\theta-\int\sec\theta\cdot(\sec^2\theta-1)\,\mathrm{d}\theta=\sec\theta\cdot\tan\theta-\int\sec^3\theta\,\mathrm{d}\theta+\int\sec\theta\,\mathrm{d}\theta\implies\int\sec^3\theta\,\mathrm{d}\theta=\frac{\sec\theta\cdot\tan\theta+\int\sec\theta\,\mathrm{d}\theta}{2}=\frac{\sec\theta\cdot\tan\theta+\ln|\sec\theta+\tan\theta|}{2}$; $\displaystyle\int\sqrt{x^2+a^2}\,\mathrm{d}x=a^2\int\sec^3\theta\,\mathrm{d}\theta=a^2\frac{\frac{\sqrt{x^2+a^2}}{a}\cdot\frac{x}{a}+\ln\left|\frac{\sqrt{x^2+a^2}}{a}+\frac{x}{a}\right|}{2}=\frac{x\sqrt{x^2+a^2}+a^2\ln\left|\sqrt{x^2+a^2}+x\right|-a^2\ln|a|}{2}$

5. 令 $x=a\sec\theta$, 則 $\displaystyle\int\frac{1}{\sqrt{x^2-a^2}}\,\mathrm{d}x=\int\frac{a\sec\theta\tan\theta}{a\tan\theta}\,\mathrm{d}\theta=\int\sec\theta\,\mathrm{d}\theta=\ln|\sec\theta+\tan\theta|$
$\displaystyle=\ln\left|\frac{x}{a}+\frac{\sqrt{x^2-a^2}}{a}\right|.$

6. 令 $u=\tan\dfrac{x}{2}$, 則 $\displaystyle\int\frac{1}{a+\sin x}\,\mathrm{d}x=\int\frac{1}{a+\frac{2u}{1+u^2}}\frac{2}{1+u^2}\,\mathrm{d}u=2\int\frac{1}{au^2+2u+a}\,\mathrm{d}u=\frac{2}{a\sqrt{\frac{a^2-1}{a^2}}}\tan^{-1}\frac{u+\frac{1}{a}}{\sqrt{\frac{a^2-1}{a^2}}}$
$\displaystyle=\frac{2}{\sqrt{a^2-1}}\tan^{-1}\frac{au+1}{\sqrt{a^2-1}}=\frac{2}{\sqrt{a^2-1}}\tan^{-1}\frac{a\tan\frac{x}{2}+1}{\sqrt{a^2-1}}.$

4.4 瑕積分

定義 (瑕積分 (improper integral)).

- 無限區間 (第一型) 瑕積分
  - 若 $f(x)$ 在 $[a, \infty)$ 連續, 則 $\displaystyle\int_a^\infty f(x)\,\mathrm{d}x = \lim_{b\to\infty}\int_a^b f(x)\,\mathrm{d}x$.
  - 若 $f(x)$ 在 $(-\infty, b]$ 連續, 則 $\displaystyle\int_{-\infty}^b f(x)\,\mathrm{d}x = \lim_{a\to-\infty}\int_a^b f(x)\,\mathrm{d}x$.
  - 若 $f(x)$ 在 $(-\infty, \infty)$ 連續, 則任取 $c \in \mathbb{R}$, $\displaystyle\int_{-\infty}^\infty f(x)\,\mathrm{d}x = \int_{-\infty}^c f(x)\,\mathrm{d}x + \int_c^\infty f(x)\,\mathrm{d}x$.
- 不連續點 (第二型) 瑕積分
  - 若 $f(x)$ 在 $(a, b]$ 連續, 則 $\displaystyle\int_a^\infty f(x)\,\mathrm{d}x = \lim_{c\to a+}\int_c^b f(x)\,\mathrm{d}x$.
  - 若 $f(x)$ 在 $[a, b)$ 連續, 則 $\displaystyle\int_a^b f(x)\,\mathrm{d}x = \lim_{c\to b-}\int_a^c f(x)\,\mathrm{d}x$.
  - 令 $c \in (a, b)$. 若 $f(x)$ 在 $[a, c) \cup (c, b]$ 連續且在 $x = c$ 不連續, 則 $\displaystyle\int_a^b f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x$.

例. 求 $\displaystyle\int_{-\infty}^\infty \frac{1}{1+x^2}\,\mathrm{d}x$.

解. $\displaystyle\int_{-\infty}^\infty \frac{1}{1+x^2}\,\mathrm{d}x = 2\int_0^\infty \frac{1}{1+x^2}\,\mathrm{d}x = 2\lim_{b\to\infty}\int_0^b \frac{1}{1+x^2}\,\mathrm{d}x = 2\lim_{b\to\infty}\tan^{-1} b = 2\cdot\frac{\pi}{2} = \pi$.

例. 求 $\displaystyle\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x$.

解. $\displaystyle\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x = \lim_{b\to\infty}\int_1^b \frac{1}{x^2}\,\mathrm{d}x = \lim_{b\to\infty}\big(-\frac{1}{b}+1\big) = 1$

例. 求 $\displaystyle\int_0^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x$.

解. $\displaystyle\int_0^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x = \lim_{c\to 0+}\int_c^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x = \lim_{c\to 0+} 2\sqrt{x}\Big|_c^1 = \lim_{c\to 0+}(2-2\sqrt{c}) = 2$

定理. 給定 $0 < a < \infty$.

- $\displaystyle\int_a^\infty \frac{1}{x^p}\,\mathrm{d}x$ 當 $p > 1$ 收斂至 $\dfrac{a^{1-p}}{p-1}$, 當 $p \leqslant 1$ 發散至 ∞.
- $\displaystyle\int_0^a \frac{1}{x^p}\,\mathrm{d}x$ 當 $p < 1$ 收斂至 $\dfrac{a^{1-p}}{1-p}$, 當 $p \geqslant 1$ 發散至 ∞.

定理. 令 $-\infty \leqslant a < b \leqslant \infty$, f, g 在 (a, b) 連續, 且 $0 \leqslant f(x) \leqslant g(x)\ \forall\, x$.

- 若 $\displaystyle\int_a^b g(x)\,\mathrm{d}x$ 收斂, $\displaystyle\int_a^b f(x)\,\mathrm{d}x$ 收斂.
- 若 $\displaystyle\int_a^b f(x)\,\mathrm{d}x$ 發散, $\displaystyle\int_a^b g(x)\,\mathrm{d}x$ 發散.

定理. 令 f, g 在 $[a, \infty)$, $a \in \mathbb{R}$ 連續, 均為正值, 且 $\lim\limits_{x\to\infty} \dfrac{f(x)}{g(x)}$ 存在, 則 $\displaystyle\int_a^\infty f(x)\,\mathrm{d}x$ 與 $\displaystyle\int_a^\infty g(x)\,\mathrm{d}x$ 同為收斂或發散.

例. 證明 $\displaystyle\int_0^\infty e^{-x^2}\,\mathrm{d}x$ 收斂.

解. 由 $e^{-x^2} \leqslant 1\ \forall 0 \leqslant x < 1$ 及 $e^{-x^2} \leqslant e^{-x}\ \forall x \geqslant 1$, $\displaystyle\int_0^\infty e^{-x^2}\,\mathrm{d}x = \int_0^1 e^{-x^2}\,\mathrm{d}x + \int_1^\infty e^{-x^2}\,\mathrm{d}x \leqslant \int_0^1 1\,\mathrm{d}x + \int_1^\infty e^{-x}\,\mathrm{d}x = 1 + \frac{1}{e}$, 故 $\displaystyle\int_0^\infty e^{-x^2}\,\mathrm{d}x$ 收斂.

例 (Γ 函數)**.** 已知 $\displaystyle\int_0^\infty e^{-t^2}\,\mathrm{d}t = \frac{\sqrt{\pi}}{2}$.

1. 證明 $\Gamma(x) \equiv \displaystyle\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂, $\forall x > 0$.
2. 證明 $\Gamma(x+1) = x\,\Gamma(x)$, $\forall x > 0$.
3. 證明 $\Gamma(n+1) = n!$, $\forall n \in \mathbb{N}$.
4. 求 $\Gamma\Big(\dfrac{1}{2}\Big)$ 與 $\Gamma\Big(\dfrac{3}{2}\Big)$.
5. 求 $\displaystyle\int_0^\infty t\,e^{-\alpha t^2}\,\mathrm{d}t$ 與 $\displaystyle\int_0^\infty t^2\,e^{-\alpha t^2}\,\mathrm{d}t$, $\alpha > 0$.

解.

1. (證一) 由 $\lim\limits_{t\to\infty} t^{x-1}e^{-\frac{t}{2}} = 0$, $\exists T > 0$ 使 $t^{x-1}e^{-\frac{t}{2}} \leqslant 1\ \forall t \geqslant T$. $\displaystyle\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t = \int_0^T t^{x-1}e^{-t}\,\mathrm{d}t + \int_T^\infty t^{x-1}e^{-t}\,\mathrm{d}t \leqslant \int_0^T t^{x-1}\,\mathrm{d}t + \int_T^\infty 1\cdot e^{-\frac{t}{2}}\,\mathrm{d}t = \frac{T^x}{x} + 2\,e^{-\frac{T}{2}}$. 故 $\forall x > 0$, $\displaystyle\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂.

   (證二) $\displaystyle\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t = \int_0^1 t^{x-1}e^{-t}\,\mathrm{d}t + \int_1^\infty t^{x-1}e^{-t}\,\mathrm{d}t$. $\displaystyle\int_1^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂, 因為

   - $\displaystyle\int_0^1 t^{x-1}e^{-t}\,\mathrm{d}t \leqslant \int_0^1 t^{x-1}\,\mathrm{d}t = \frac{t^x}{x}\Big|_{x=0}^{x=1} = \frac{1}{x}$, $\displaystyle\int_0^1 t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂.
   - $\displaystyle\int_1^\infty \frac{1}{t^2}\,\mathrm{d}t$ 收斂, $\lim\limits_{t\to\infty} \dfrac{t^{x-1}e^{-t}}{\frac{1}{t^2}} = \lim\limits_{t\to\infty} \dfrac{t^{x+1}}{e^t} = 0$, $\displaystyle\int_1^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂.

2. $\displaystyle\Gamma(x+1) = \int_0^\infty t^x e^{-t}\,\mathrm{d}t = \lim_{\substack{a\to 0+\\ b\to\infty}} \int_a^b t^x e^{-t}\,\mathrm{d}t$. 令 $u = t^x$, 則 $\mathrm{d}u = xt^{x-1}\,\mathrm{d}t$. 令 $\mathrm{d}v = e^{-t}\,\mathrm{d}t$, 則 $v = -e^{-t}$.
   故 $\displaystyle\lim_{\substack{a\to 0+\\ b\to\infty}} \int_a^b t^x e^{-t}\,\mathrm{d}t = \lim_{\substack{a\to 0+\\ b\to\infty}} \Big(-t^x e^{-t}\Big|_a^b + \int_a^b e^{-t}\cdot xt^{x-1}\,\mathrm{d}t\Big) = 0 + x\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t = x\,\Gamma(x)$.

3. 由上 $\Gamma(n+1) = n\,\Gamma(n) = n(n-1)\,\Gamma(n-2) = \cdots = n(n-1)\cdots 2\,\Gamma(1)$, 又 $\Gamma(1) = \displaystyle\lim_{b\to\infty} \int_0^b e^{-t}\,\mathrm{d}t = \lim_{b\to\infty} -e^{-t}\Big|_0^b = \lim_{b\to\infty} 1 - e^{-b} = 1$, 得證.

4. (a) $\displaystyle\Gamma\Big(\frac{1}{2}\Big) = \int_0^\infty t^{-\frac{1}{2}}e^{-t}\,\mathrm{d}t = \int_0^\infty \frac{e^{-t}}{\sqrt{t}}\,\mathrm{d}t$. 令 $u = \sqrt{t}$, 則 $t = u^2$, $\mathrm{d}u = \dfrac{1}{2\sqrt{t}}\,\mathrm{d}t \implies \dfrac{1}{\sqrt{t}}\,\mathrm{d}t = 2\,\mathrm{d}u$.
   積分範圍 t 由 0 至 ∞, 則變數變換後 u 由 $\sqrt{0} = 0$ 至 $\sqrt{\infty} = \infty$, 故 $\displaystyle\Gamma\Big(\frac{1}{2}\Big) = \int_0^\infty \frac{e^{-t}}{\sqrt{t}}\,\mathrm{d}t = 2\int_0^\infty e^{-u^2}\,\mathrm{d}u = 2\cdot\frac{\sqrt{\pi}}{2} = \sqrt{\pi}$.

   (b) 由 $\Gamma(x+1) = x\,\Gamma(x)$, $\Gamma\Big(\dfrac{3}{2}\Big) = \Gamma\Big(\dfrac{1}{2}+1\Big) = \dfrac{1}{2}\,\Gamma\Big(\dfrac{1}{2}\Big) = \dfrac{1}{2}\sqrt{\pi}$.

5. (a) 令 $u = \alpha\, t^2$, 則 $\mathrm{d}u = 2\,\alpha\, t\,\mathrm{d}t \implies t\,\mathrm{d}t = \dfrac{\mathrm{d}u}{2\alpha}$. 積分範圍 t 由 0 至 ∞, 則變數變換後 u 由 $\alpha \cdot 0^2 = 0$ 至 $\alpha \cdot \infty^2 = \infty$, 故 $\displaystyle\int_0^\infty t\, e^{-\alpha\, t^2}\,\mathrm{d}t = \int_0^\infty e^{-u}\,\frac{\mathrm{d}u}{2\,\alpha} = \frac{1}{2\,\alpha}\int_0^\infty e^{-u}\,\mathrm{d}u = \frac{1}{2\,\alpha}\Big(-e^{-u}\Big|_0^\infty\Big) = \frac{1}{2\,\alpha}$.

(b) 令 $u = \alpha\, t^2$, 則 $t = \dfrac{\sqrt{u}}{\sqrt{\alpha}}$, $\mathrm{d}u = 2\,\alpha\, t\,\mathrm{d}t \implies t\,\mathrm{d}t = \dfrac{\mathrm{d}u}{2\,\alpha}$. 積分範圍 t 由 0 至 ∞, 則變數變換後 u 由 $\alpha \cdot 0^2 = 0$ 至 $\alpha \cdot \infty^2 = \infty$, 故 $\displaystyle\int_0^\infty t^2\, e^{-\alpha\, t^2}\,\mathrm{d}t = \int_0^\infty t\, e^{-\alpha\, t^2}\, t\,\mathrm{d}t = \int_0^\infty \frac{\sqrt{u}}{\sqrt{\alpha}}\, e^{-u}\,\frac{\mathrm{d}u}{2\,\alpha} = \frac{1}{2\,\alpha^{\frac{3}{2}}}\int_0^\infty u^{\frac{1}{2}}\, e^{-u}\,\mathrm{d}u = \frac{1}{2\,\alpha^{\frac{3}{2}}}\cdot\Gamma\Big(\frac{3}{2}\Big) = \frac{1}{2\,\alpha^{\frac{3}{2}}}\cdot\frac{1}{2}\sqrt{\pi} = \frac{\sqrt{\pi}}{4\,\alpha^{\frac{3}{2}}}$.